Digital Twins

Thirtieth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Digital Twin in Manufacturing: Challenges and Roadmap

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From tools to machine tools







Turning





Milling

Drilling



ECM



Can we replace the operator with machine?

- Decision support system
- Learning with experience
- Self-correction ability





Industry 4.0 - Overview

- 1st Industrial Revolution (Later half of 18th century) – The use of steam power.
- 2nd Industrial Revolution (Later part of 19th century) – The use of electrical energy, mass production.
- 3rd Industrial Revolution (Later part of 20th century) – The use of electronics and use of robotics in manufacturing.
- 4th Industrial Revolution (Present time) – The use of cyber physical system for controlling machines with little human intervention, digital manufacturing

The large scale innovation in technology "*pushed*" which the market demand "*pulled*" for a long time leading to the emergence of German coined "*Industrie 4.0*" (Lasi et al., 2014).



Lasi, H., Fettke, P., Kemper, H.G., Feld, T. and Hoffmann, M., 2014. Industry 4.0. *Business & information systems engineering*, *6(4),* pp.239-242.

Invention of Transistor

The transistor was invented in 1947 by John Bardeen, Walter Brattain, and William Shockley at Bell Labs.

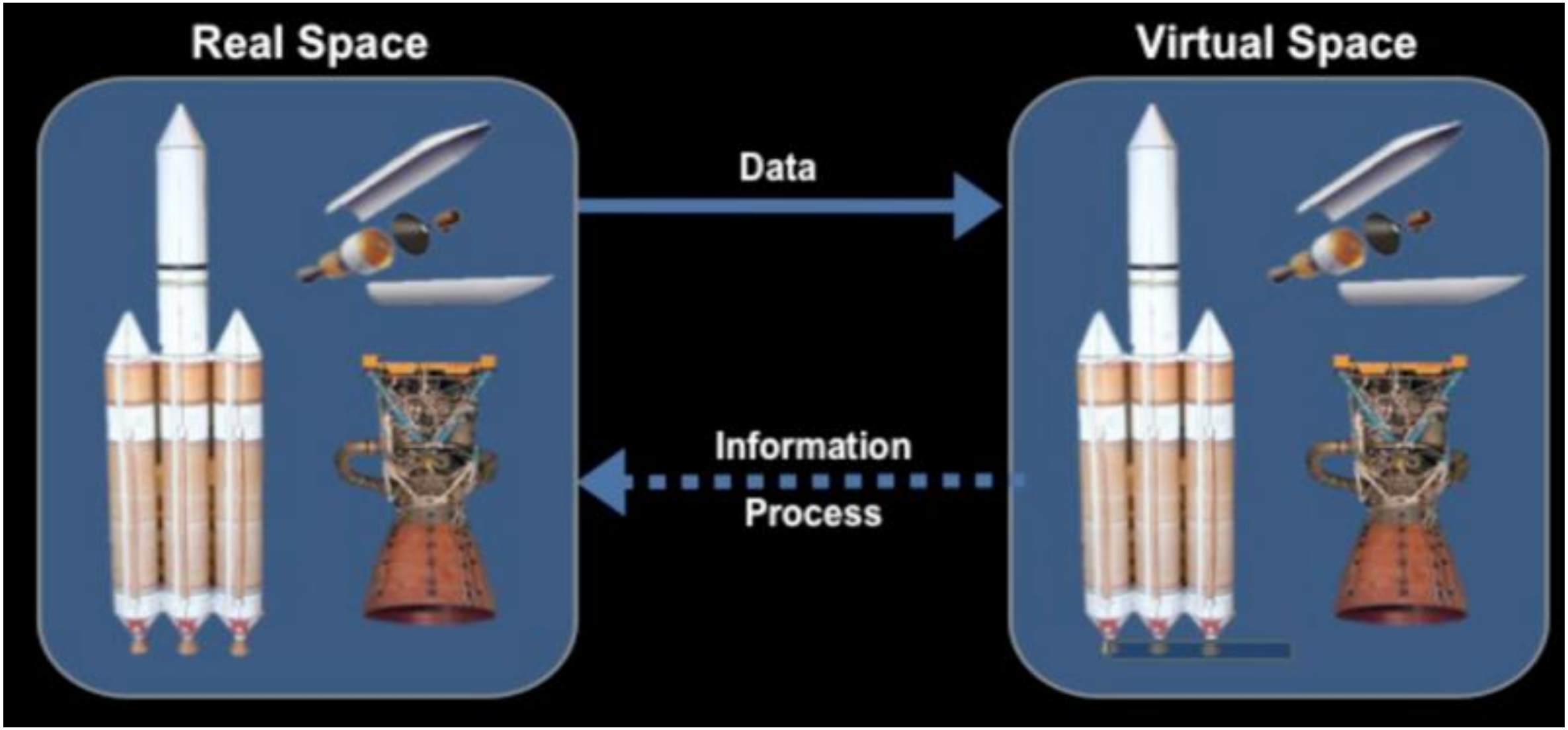
In 1956, Bardeen, Brattain, and Shockley were awarded the Nobel Prize in Physics for their work.



AT&T; photographer: Jack St.

Digital Twin in Manufacturing

- Digital Twin is a virtual replica of a physical model or process.
- It is mainly used for controlling, monitoring and evaluating the performance of the physical model or process under consideration.
- Real-time monitoring of the manufacturing processes using digital twin enables the manufacturers to improve efficiency and throughput.
- The ease in the availability of data captured from sensors has made it much easier to create digital twins



https://commons.wikimedia.org/wiki/File:Digital_Twin_Concept_of_Grieves_and_Vickers.png

Digital Twin

Digital Shadow

Digital Model

Digital Model

It is a digital representation of a physical object or process. Used for design, visualization or analysis. Eg: A CAD model of a car or a building.

Digital Shadow

A digital representation that receives data from the physical system. There is no feedback from the digital to the physical system. Data flows only in one direction. Only reflects the current state of the physical system but cannot reflect it.

Eg: Educational Institute bus that sends data like GPS location to cloud.

Digital Twin

A **bi-directional**, real-time, connected digital representation of a physical system. Data flows in both ways. Digital twin can simulate, analyze and control the physical system.

Eg: A wind turbine with a digital twin that monitors its performance, predicts failures and can adjust blade angles remotely.

Modelling

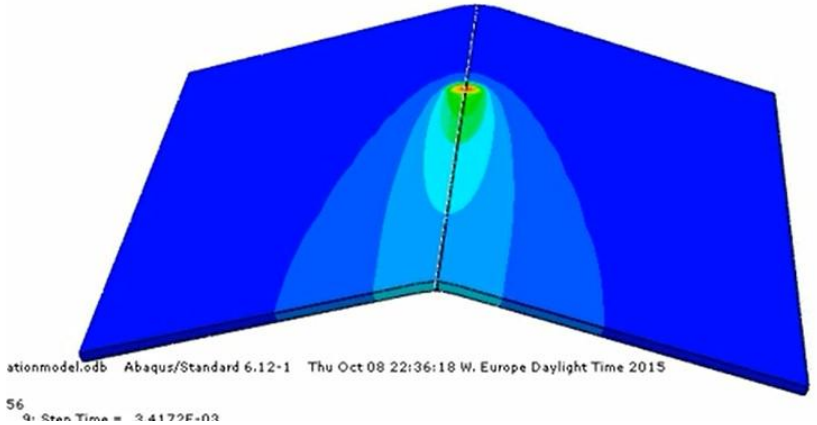
- Model is the representation of the process that provides some information about it.
- Execution of model is called simulation.
- Two types of models: Direct and Inverse

Definition of Simulation

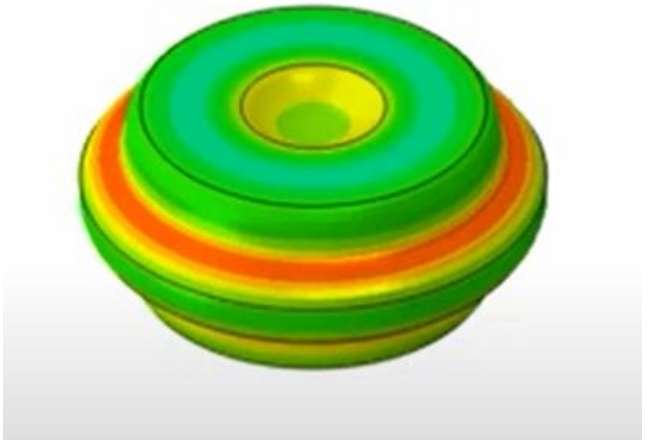
- Simulation is an approximate imitation of the process.
- Simulation is the imitative representation of the functioning of one system or process by means of the functioning of another. (Meriam-Webster)
- Simulation modelling and analysis is the process of creating and experimenting with a computerized mathematical model of a physical system. (Chung C., 2004, Simulation modelling handbook: a practical approach, 1st Ed., CRC press, Boca Raton.)

Finite Element Modelling (FEM) in ABAQUS

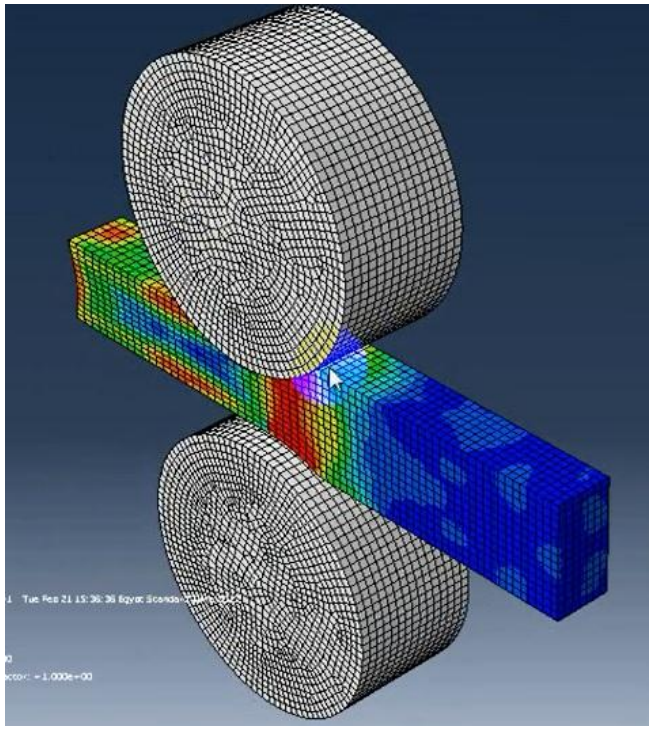
- FEM simulations using ABAQUS creates a virtual copy of the actual physical model for detailed analysis.
- It can be considered as an example of digital model.
- ABAQUS simulations closely resembles an actual physical system.



Laser Bending Simulation



Closed-die Forging Simulation



Cold Rolling Simulation

MD vs CP FEM

MD:

- ✓ Useful when underlying deformation mechanism not well understood
- ✓ Gives fundamental insight of the deformation processes
- ✓ No prior knowledge of system evolution required

- ○ Increased complexity of the potential and computational cost for multi-component system
- ○ Timescale (~ns) and domain size (~100 nm) are very limited
- ○ Unrealistic strain rate (10^8 s^{-1} and above) far from real-world situation simulated

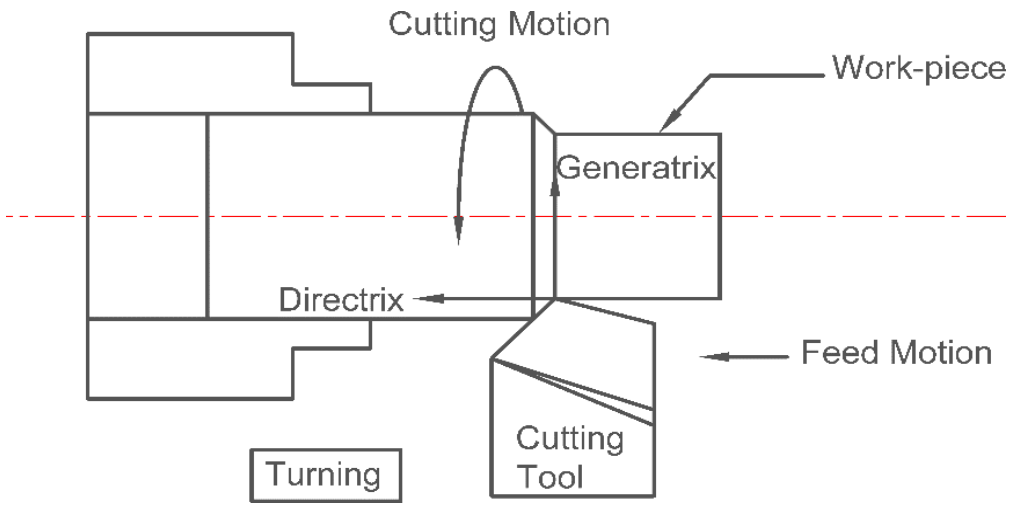
CP FEM:

- ✓ Computationally more efficient
- ✓ Capability to model reasonable spatial and time domains
- ✓ Various deformation modes (slip, twin, etc) can be modelled
- ✓ Material with different crystal structure (FCC, HCP, BCC) can be modelled
- ✓ Deformation mode dependency on various parameters (eg. Size, temperature, strain rate, etc) can be incorporated
- ✓ Capability to investigate individual deformation mode (separately)

- ○ Phenomenological material laws need to be established for new material via initial experimental investigation
- ○ Involves identification of deformation modes and calibration of material model

Issues in the tradition method of modelling a manufacturing process

- Machining is one of the widely used manufacturing process.
- Still, it is one of the least understood process in manufacturing (Usui, 1988).
- It is challenging to model such a complex process by analytical means.



Turning Operation

Usui, E., 1988. Progress of "predictive" theories in metal cutting. *JSME international journal. Ser. 3, Vibration, control engineering, engineering for industry*, *31*(2), pp.363-369.

Issues in the tradition method of modelling a manufacturing process (cont.)

- The traditional Merchant and Ernst (1941) model failed to explain the machining behaviour of AISI 1045 medium carbon steel and AISI 316 stainless steel (Astakhov and Outeiro, 2008).
- It is observed that despite AISI 1045 medium carbon steel having higher yield and ultimate strength than AISI 316 stainless steel, still the former experiences lower cutting forces.
- One of the reasons stated by the researchers (Astakhov and Outeiro, 2008) is Merchant's theories mainly utilises the stresses associated with metal cutting and little or no focus is on the strains associated.

Astakhov, V.P. and Outeiro, J.C., 2008. Metal cutting mechanics, finite element modelling. In Machining (pp. 1-27). Springer, London.

Issues in the tradition method of modelling a manufacturing process (cont.)

- Astakhov (2011) pointed out the following two major issues faced by the researchers while modelling a machining process:

I. Strain rates are believed to be high during machining (around 10^5 s^{-1}) (Freudenthal, 1950; Kronenberg, 1966; Stephenson and Agapiou, 1996). However, the decrease of cutting force with cutting speed and absence of twinning deformation suggest that strain cannot exceed 10 s^{-1}.

II. Cutting temperatures are considered to be high leading to material softening. However, considering the fact that most of the heat goes away with chips, it may be prudent enough to consider machining as a cold working process.

Astakhov, V.P., 2011. Authentication of FEM in metal cutting. *Finite element method in manufacturing processes*, pp.1-43
Freudenthal, A.M., 1950. *The inelastic behavior of engineering materials and structures*. Wiley
Kronenberg, M., Machining science and application 1966. *Pergamon Press, London,*, *217*, p.219.
Stephenson, D.A. and Agapiou, J.S., 2016. *Metal cutting theory and practice*. CRC press

Need for ML in the current manufacturing scenario

- The following are the major goals of the manufacturer (Ji et al., 2019):
  - Higher machining efficiency
  - Lower cost
  - Higher quality of the product
- The main aim of implementing ML in modern day industries is to ensure a cost efficient running system compromising safety (Yin and Kaynak, 2015).

Ji, W., Yin, S. and Wang, L., 2019. A big data analytics based machining optimisation approach. *Journal of Intelligent Manufacturing*, *30*(3), pp.1483-1495.

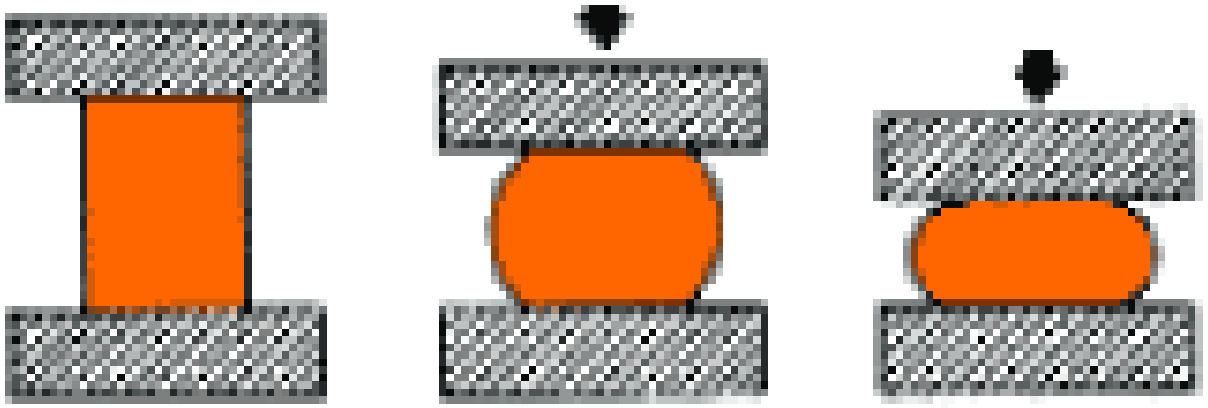
Yin, S. and Kaynak, O., 2015. Big data for modern industry: challenges and trends [point of view]. *Proceedings of the IEEE*, *103*(2), pp.143-146.

Need for ML in the current manufacturing scenario (cont.)

- The key performance indicators in machining such as cutting force, cutting tool temperature, surface quality can be directly analysed using past data as well as real time data based on the controllable and uncontrollable variables.

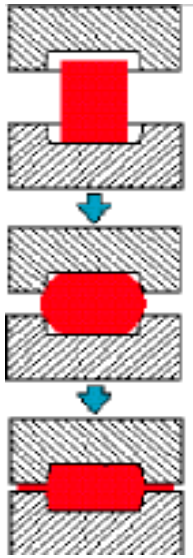
- Proper implementation of big data in manufacturing can drastically reduce the product development time and other cost associated with it as reported in a study conducted by McKinsey Global Institute Research Organization (Manyika, 2011) .

Manyika, J., 2011. Big data: The next frontier for innovation, competition, and productivity. *http://www.mckinsey.com/Insights/MGI/Research/Technology_and_Innovation/Big_data_The_next_frontier_for_innovation*..



Open-die forging

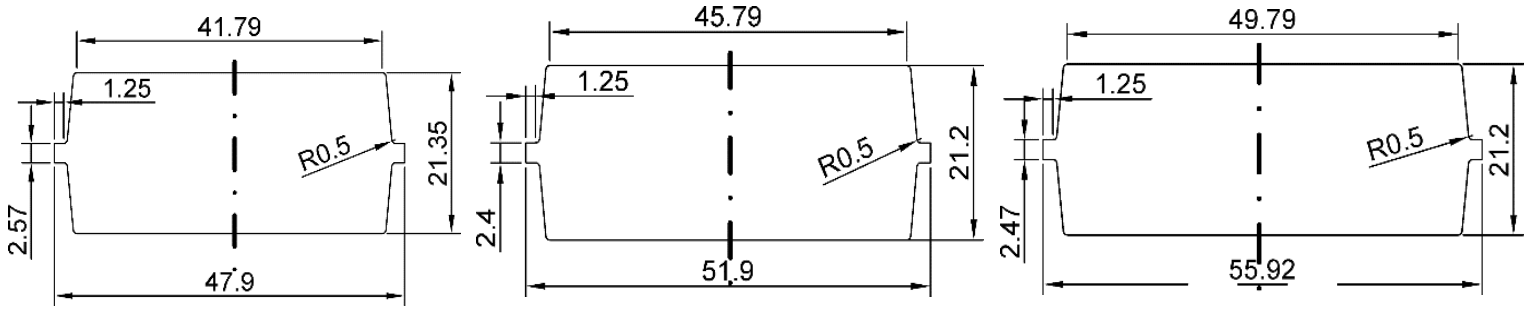
Can we predict forging load with experience?



Closed-die forging

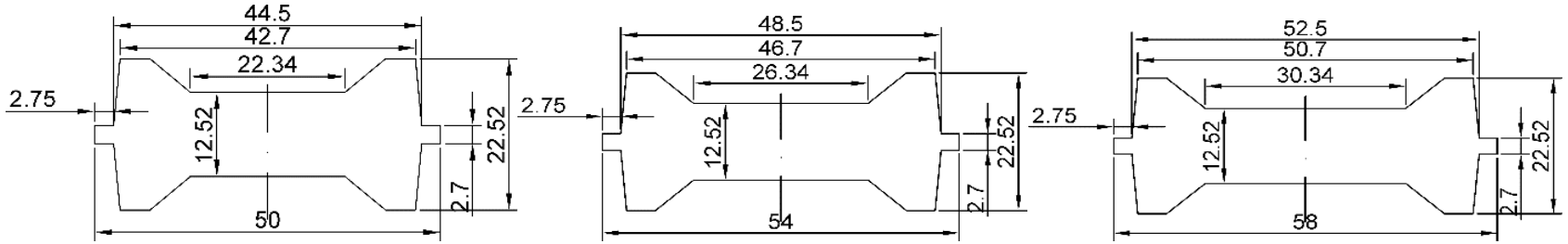
Estimating forging load in closed die forging

Forging load estimation for different sizes



(a) (b) (c)

Closed die forged Product 1: (a) Model 1, (b) Model 2 and (c) Model 3

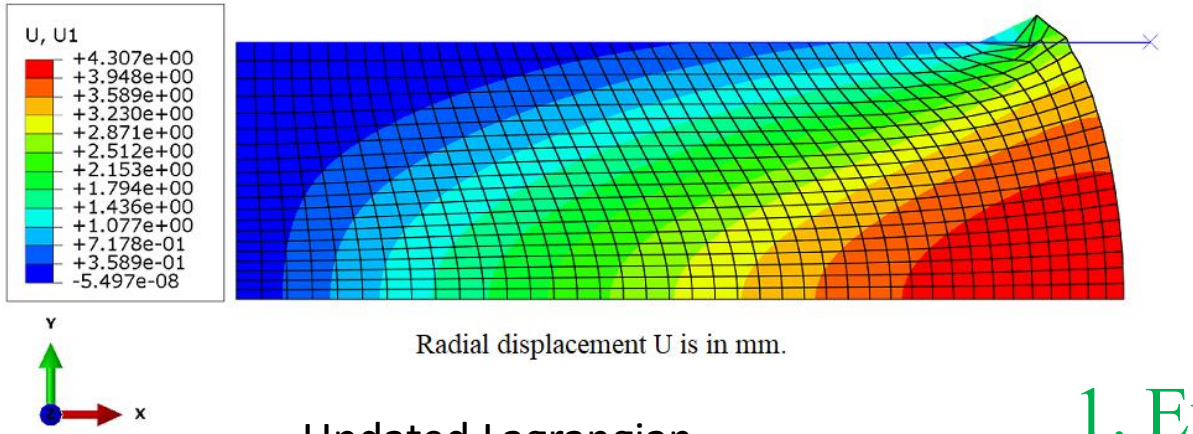


(a) (b) (c)

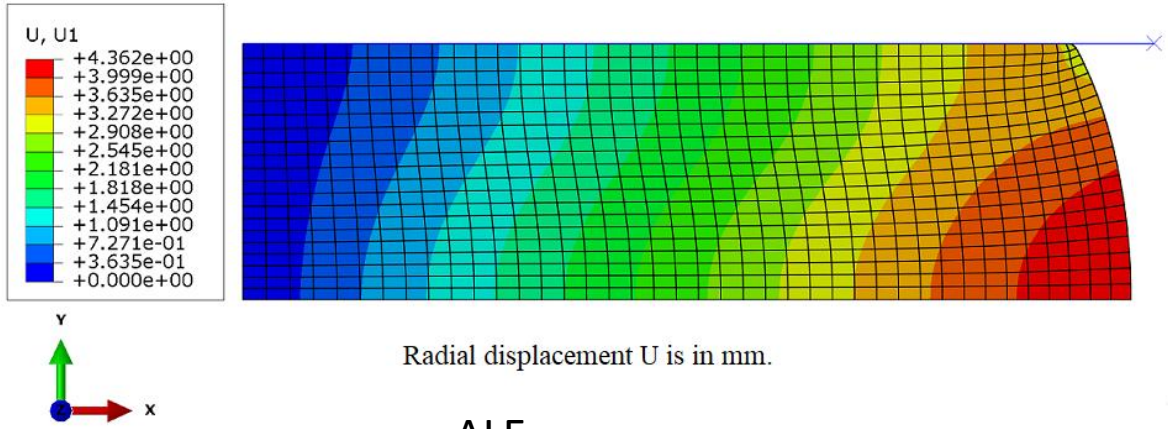
Closed die forged Product 2: (a) Model 1, (b) Model 2 and (c) Model 3.

Can machine learning help here?

Load estimation can be done by FEM also.



Updated Lagrangian



ALE

Major Problems

1. Excessive computational time
2. Uncertainty about accuracy of the model
3. Uncertainty about input parameters

What is Machine Learning (ML)?

- "The field of study that gives computers the ability to learn on their own" (Samuel, 1959).
- ML algorithms learn the pattern from past datasets and apply the same to new or unseen datasets.
- ML uses probability theory, statistics, linear algebra and calculus for drawing inferences.



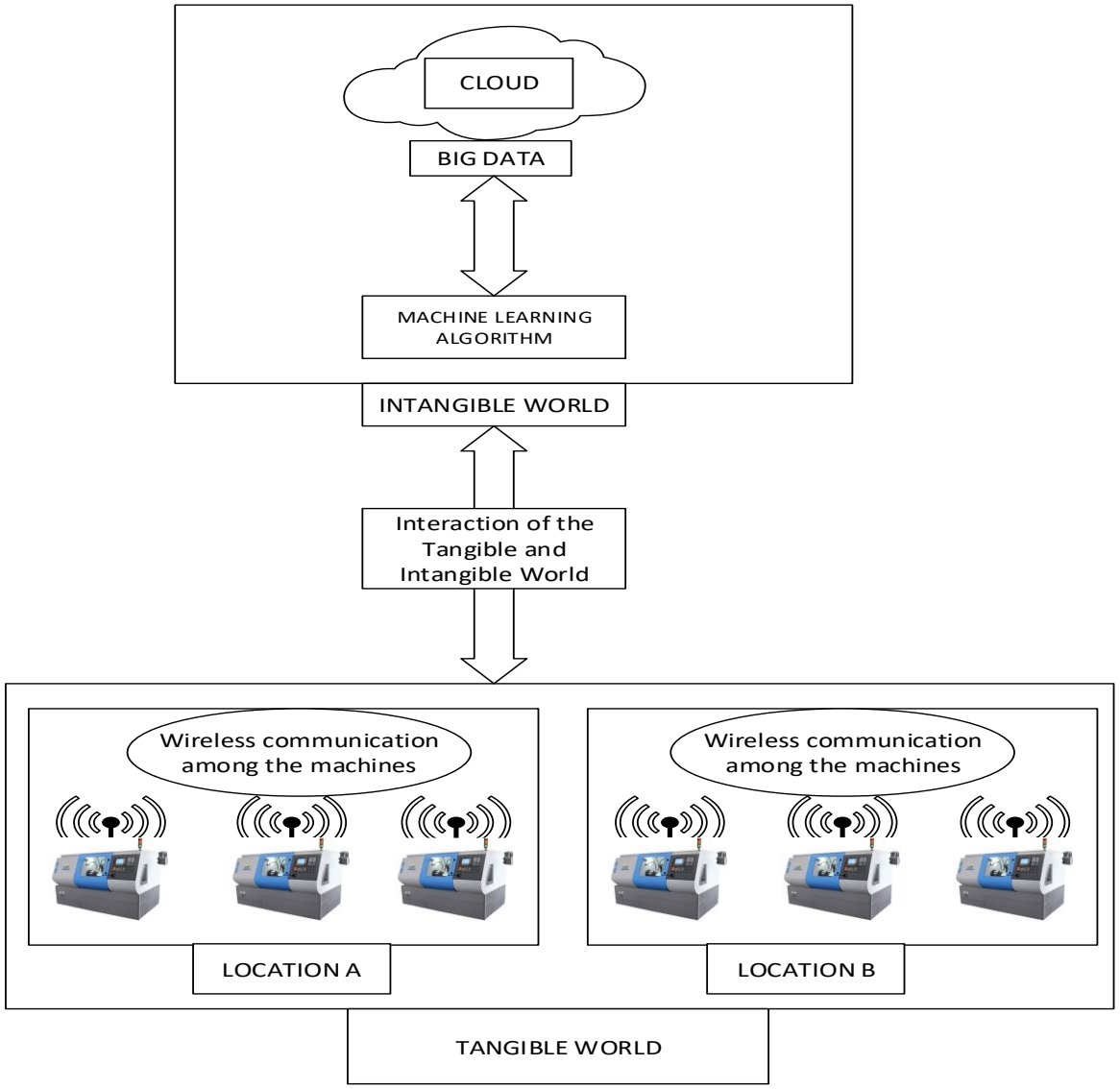
Arthur Lee Samuel

https://commons.wikimedia.org/wiki/File:This_is_the_photo_of_Arthur_Samuel.jpg

Modelling a machining process

- Machining is one of the widely used manufacturing process.
- Still it is one of the least understood process in manufacturing.
- It is difficult to model such a complex process by analytical means.
- The key performance indicators in machining such as cutting force, cutting tool temperature, surface quality can be directly analysed using past data.

Smart Manufacturing



Smart Manufacturing System

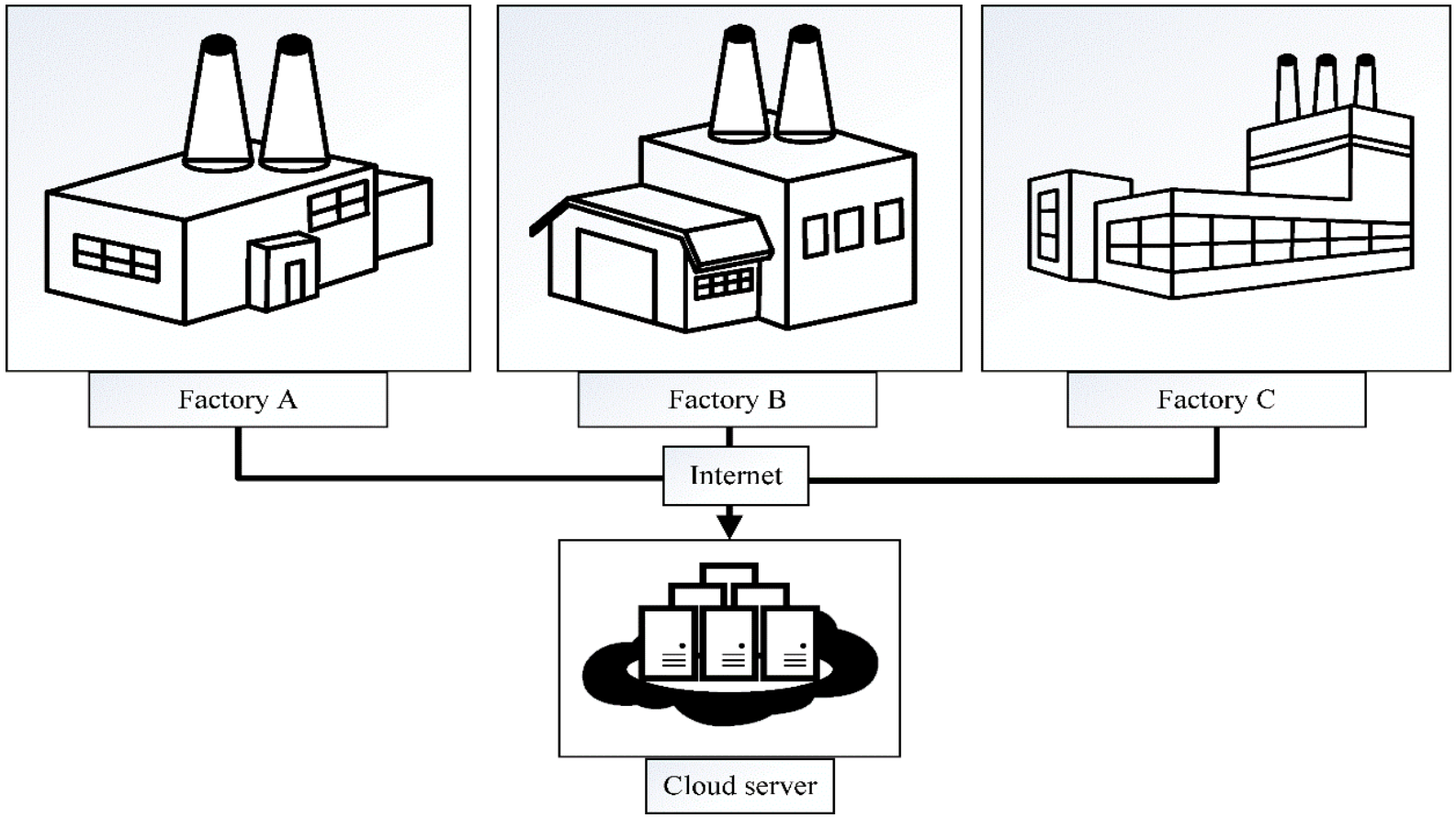
Scheme for developing and maintaining a databank

- Two examples are presented to propose a methodology on utilising a large amount of past machining data for the enhancement of the productivity.
- The first example involves collection of veracious data from various sources.
- The second example involves efficient storage and prediction.



Cloud Server

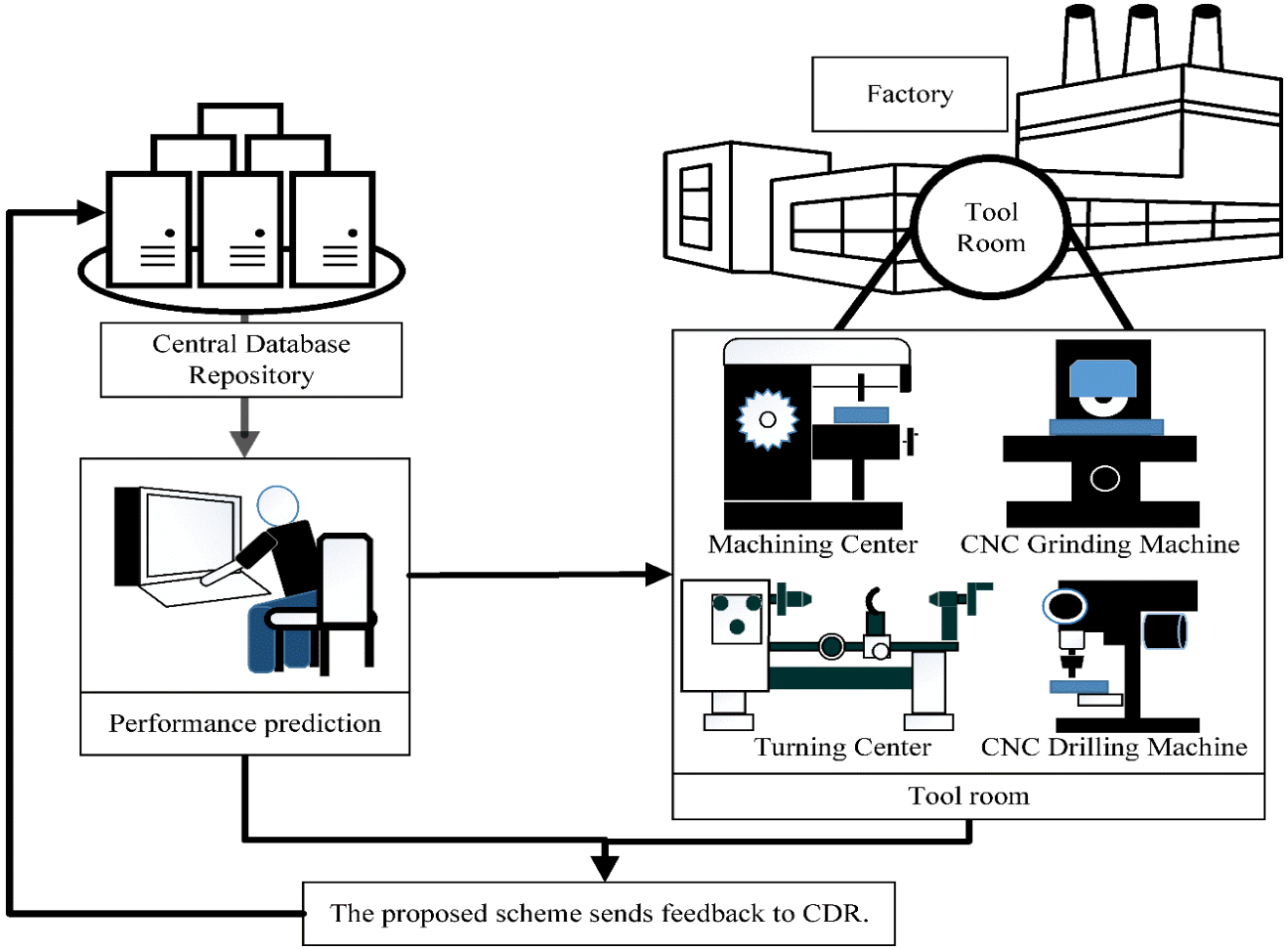
Acquisition of veracious data



Conceptual plan for sharing of data among factories

Chatterjee, K., Zhang, J. and Dixit, U.S., 2020. Data-driven framework for the prediction of cutting force in turning. IET Collaborative Intelligent Manufacturing, 2(2), pp.87-95.

Acquisition of veracious data



Proposed scheme for sending feedback to cloud based on the actual result from the shop floor

Chatterjee, K., Zhang, J. and Dixit, U.S., 2020. Data-driven framework for the prediction of cutting force in turning. IET Collaborative Intelligent Manufacturing, 2(2), pp.87-95.

Acquisition of veracious data (cont.)

- The following are the key strategies used:

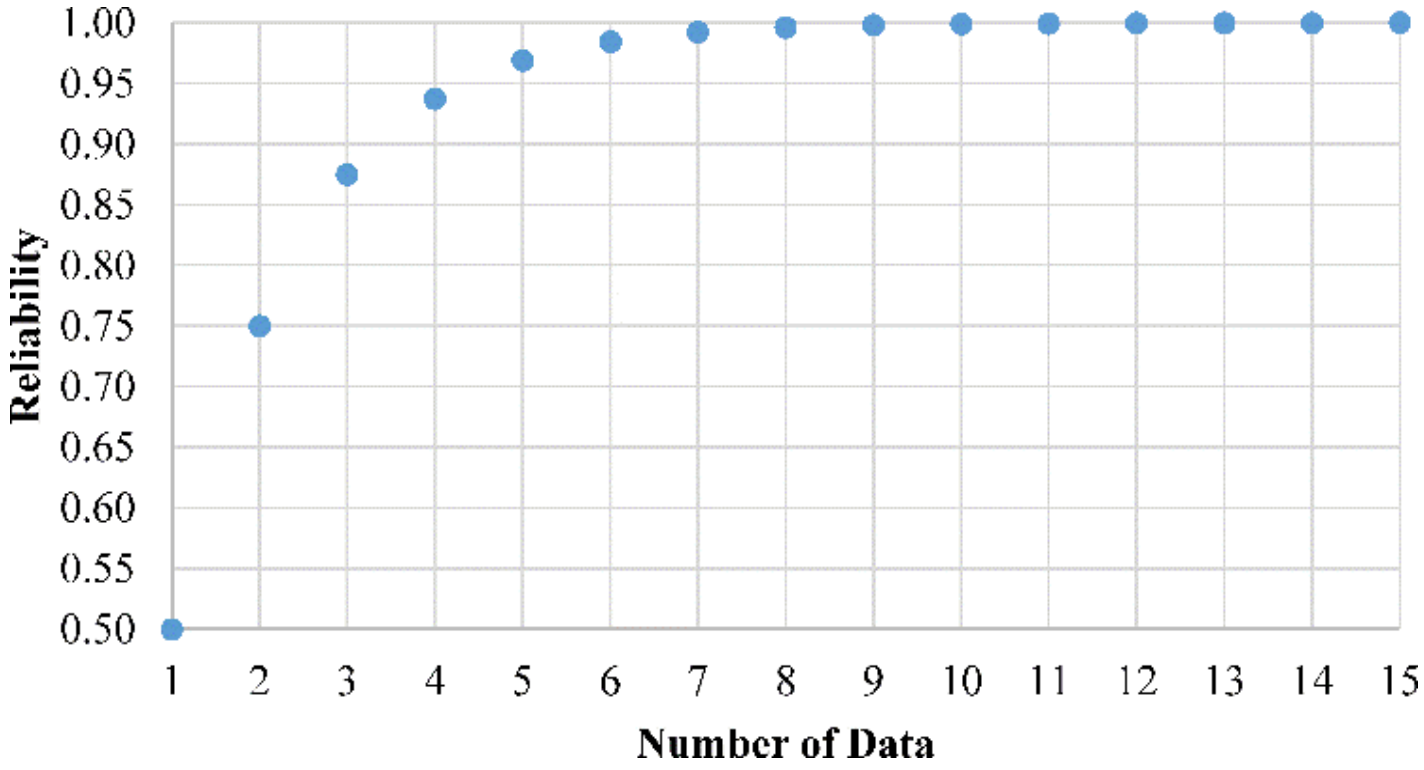
❑ Building a reliable databank for the prediction of cutting force in turning that can be shared among factories across the globe.

❑ Avoiding unnecessary data in the databank. Hence, no data explosion.

❑ Data is filtered to understand the effect of vibration, tool wear and sensor fault.

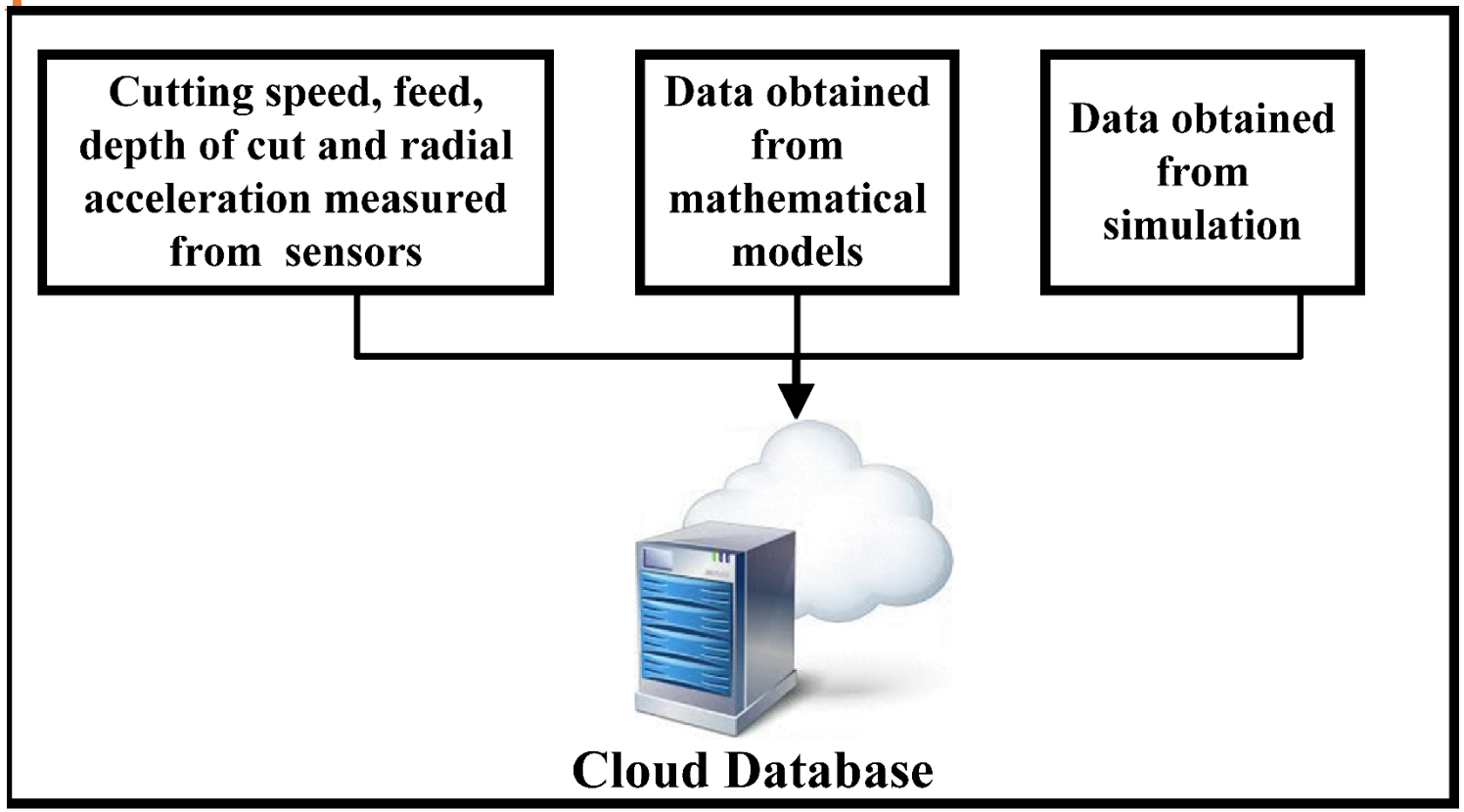
❑ A new concept of dynamic reliability is attached with every data that can provide the veracity of the information being stored.

❑ Uncertainties associated with machining are also taken care of by introducing a mini-repository for keeping and analysing such data.



Reliability increases with experience.

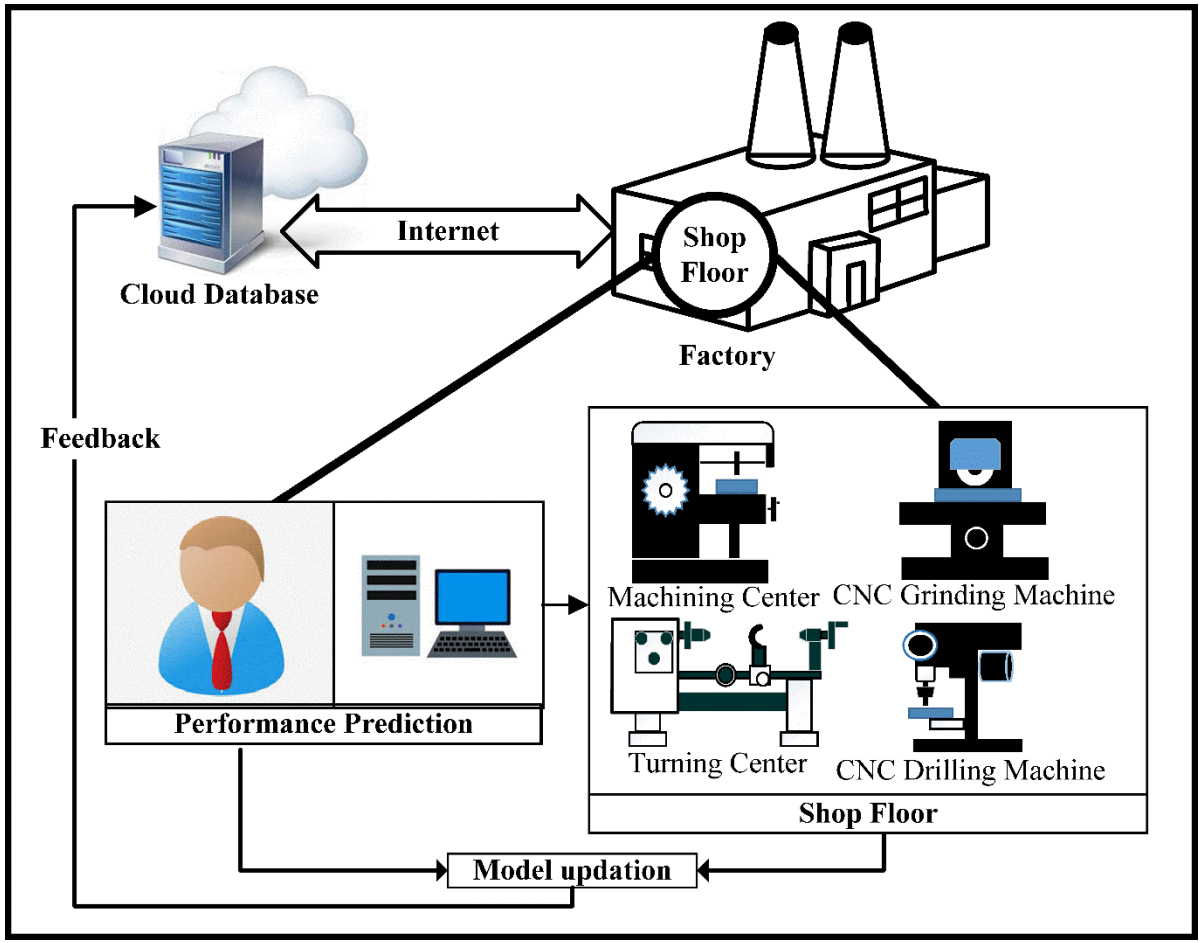
Efficient storage of data and performance prediction



Data collection module

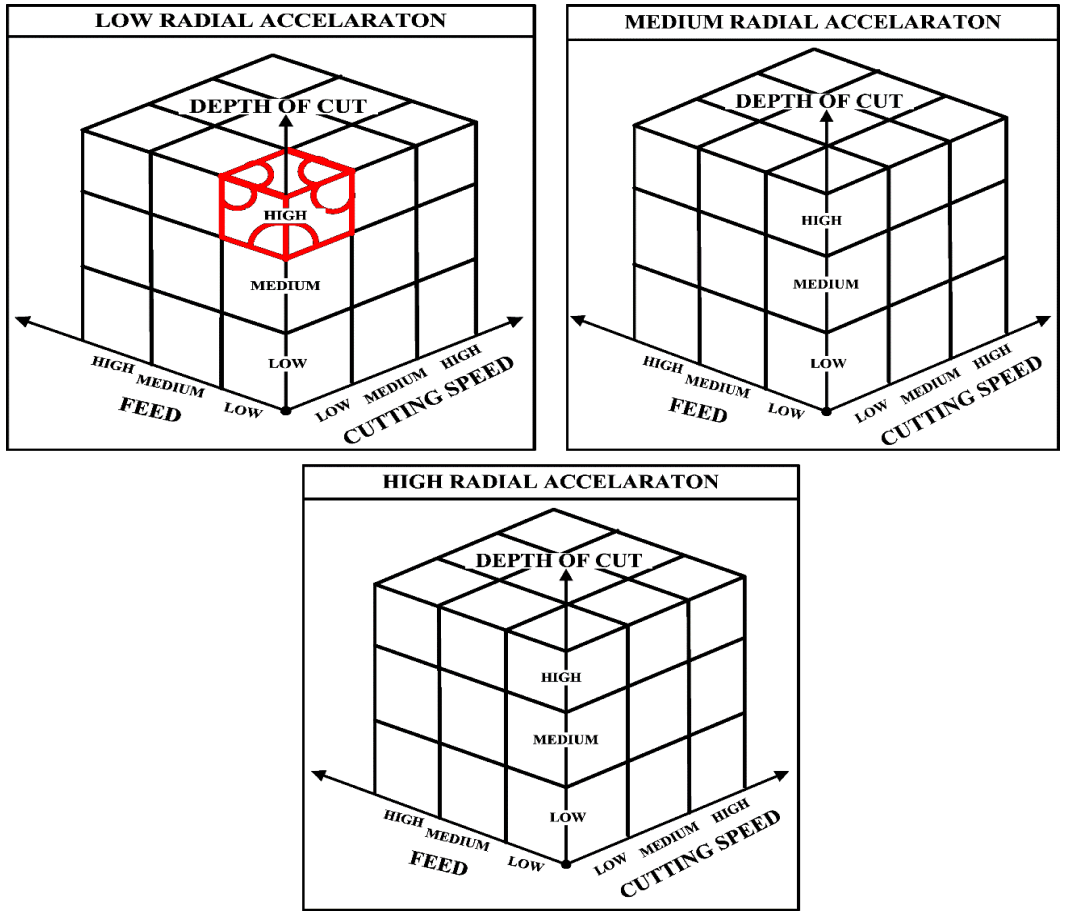
Chatterjee, K., Zhang, J. and Dixit, U.S., 2021. Estimation of surface roughness in a turning operation using industrial big data. International Journal of Machining and Machinability of Materials, 23(3), pp.209-240.

Efficient storage of data and performance prediction (cont.)



Surface roughness prediction and data updating module (Chatterjee et al., 2021)

Efficient storage of data and performance prediction (cont.)



Three cubes showing four input process parameters divided into three equal intervals (Chatterjee et al., 2021)

Efficient storage of data and performance prediction (cont.)

- The following are the key strategies used in case study 1:

❑ Case study 2 builds upon the framework shown in case study 1.

❑ Minimum number of parameters are used for the interval estimation of surface roughness in turning using 35000 datasets.

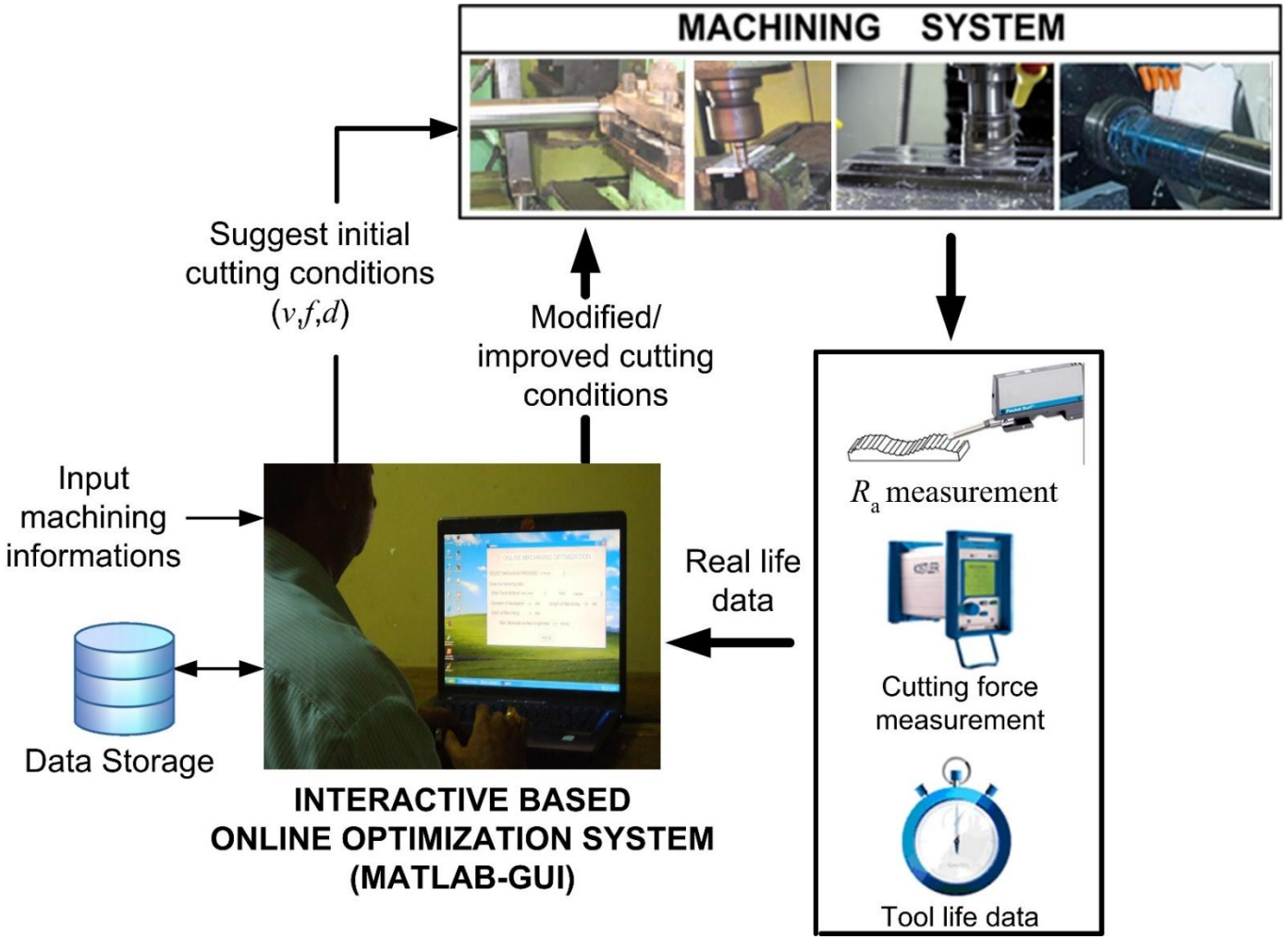
❑ The existing range of data are divided into three major levels and curve fitting is carried out.

❑ The concept of dynamic reliability is also used in this study.

❑ Influential observations are evaluated using Cook's distance and outliers are also taken of.

❑ There are provisions of updating the stored model parameters.

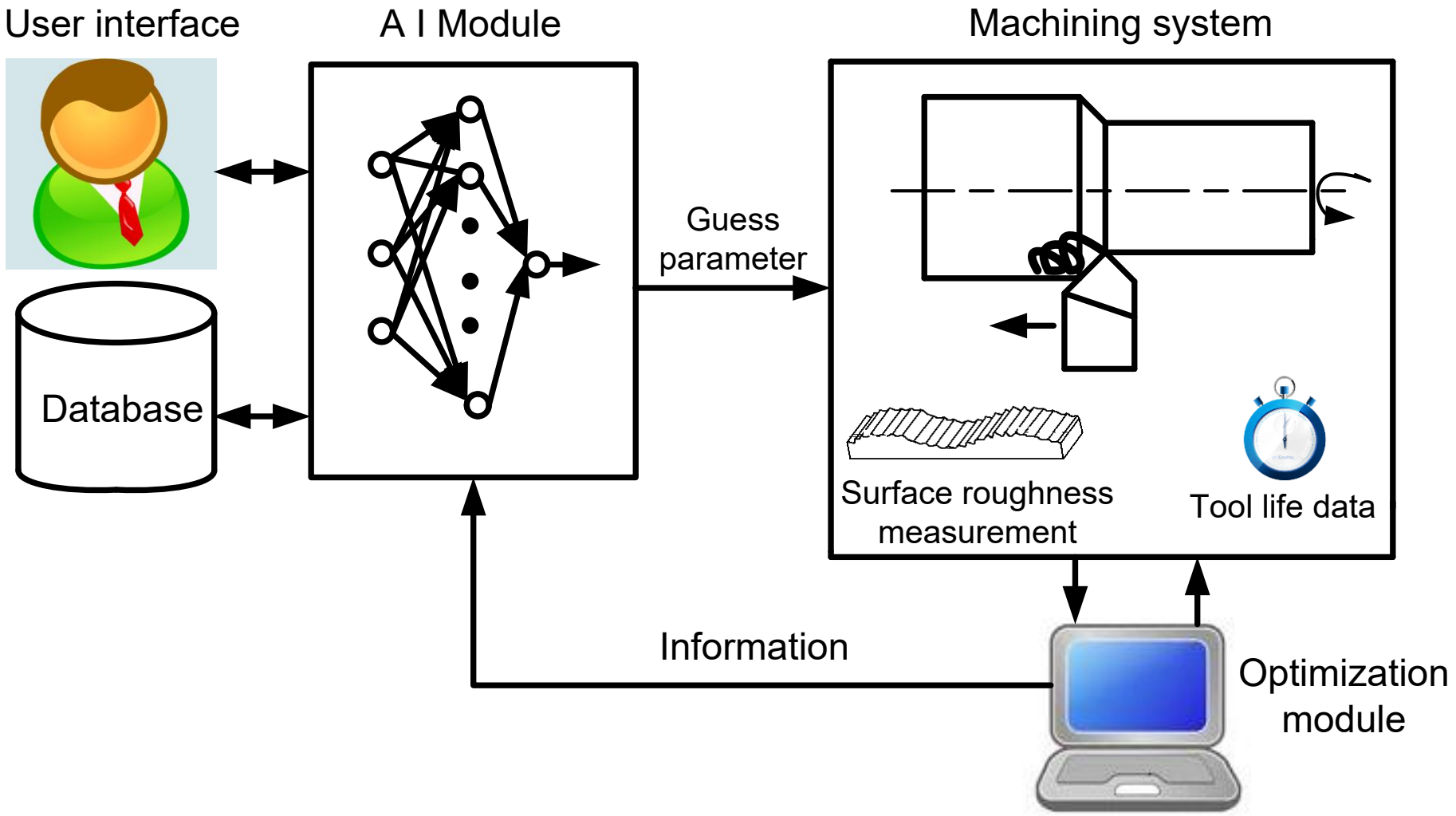
Online Optimization of Machining Process



Online optimization of machining process with continuous learning (Chandrasekaran et al., 2012)

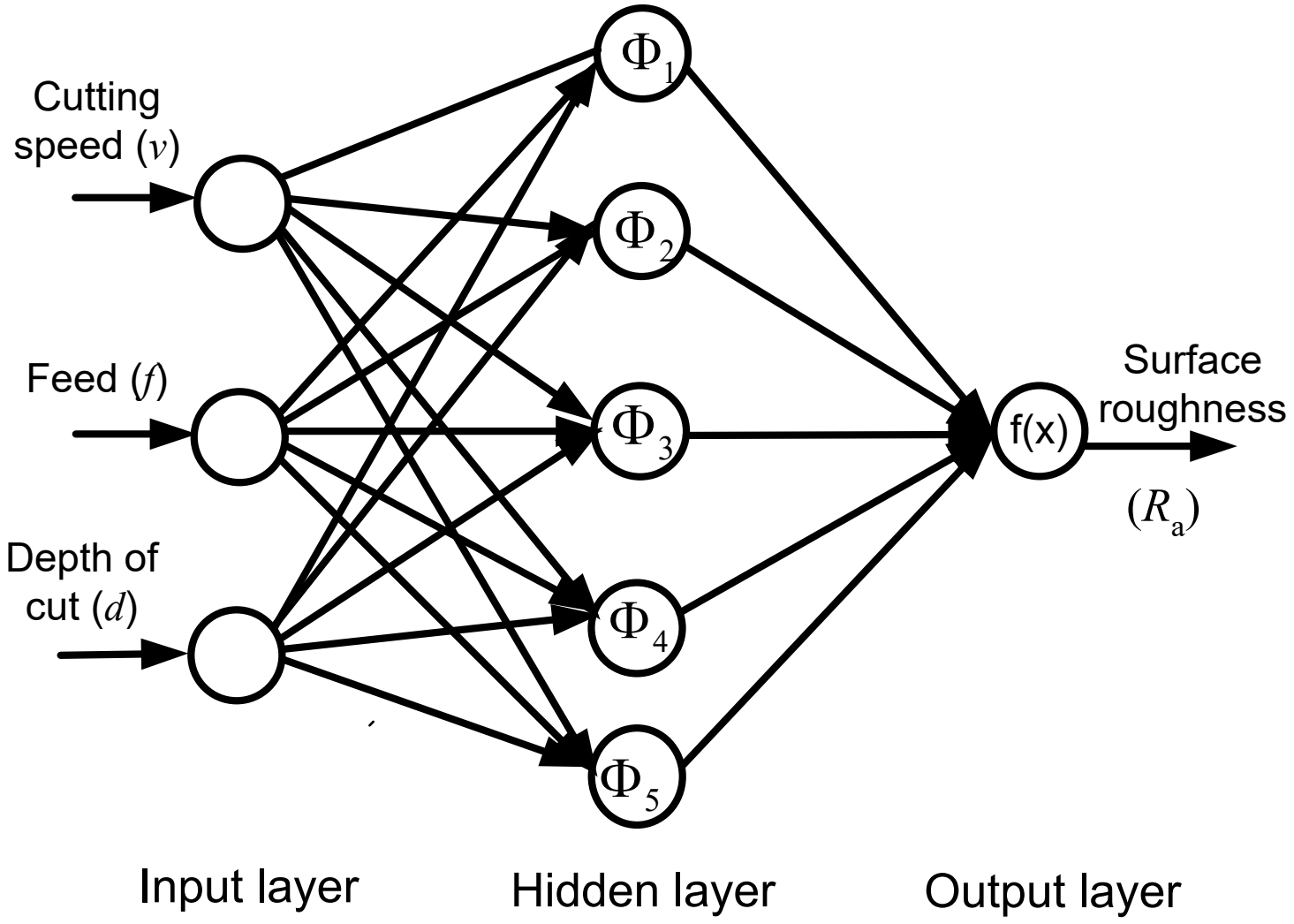
Chandrasekaran, M., Muralidhar, M., Krishna, C.M. and Dixit, U.S., 2012. Online machining optimization with continuous learning. In Computational methods for optimizing manufacturing technology: models and techniques (pp. 85-110). IGI Global.

Online Optimization of Machining Process (cont.)



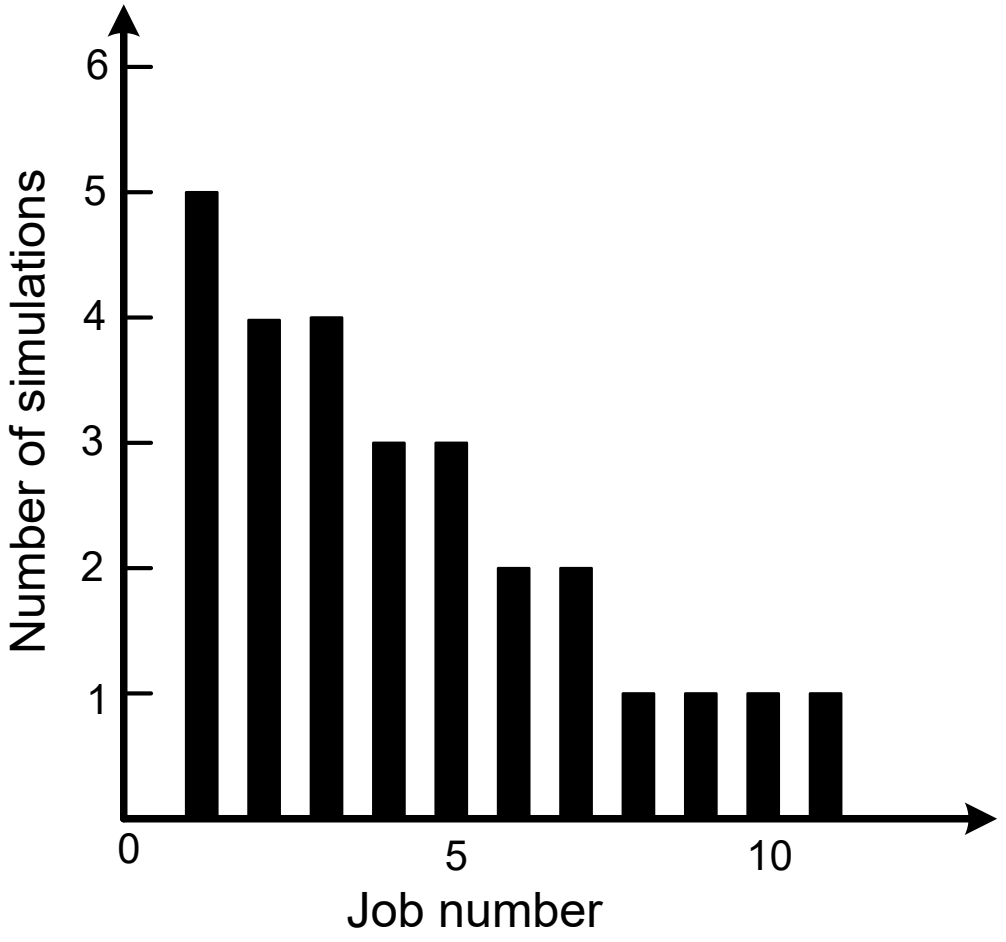
Online optimization of machining process with continuous learning (Chandrasekaran et al., 2012)

Online Optimization of Machining Process (cont.)

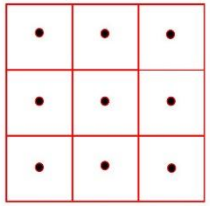


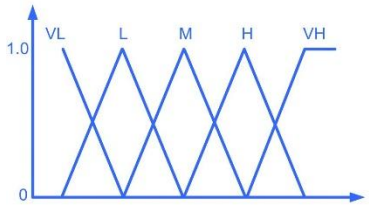
A typical neural network architecture for the AI module

Learning reduces the number of simulations

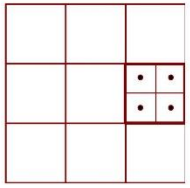


Fuzzy set based optimization





AND		y		
		L	M	H
x	L	VH	M	VH
	M	VL	L	VH
	H	VH	VH	VH



1. Divide search domain into number of cells

2. Fuzzify the decision variables in fuzzy subsets

3. Develop the rule base

4. Identify the optimum cell. Refine the search till desired accuracy obtained

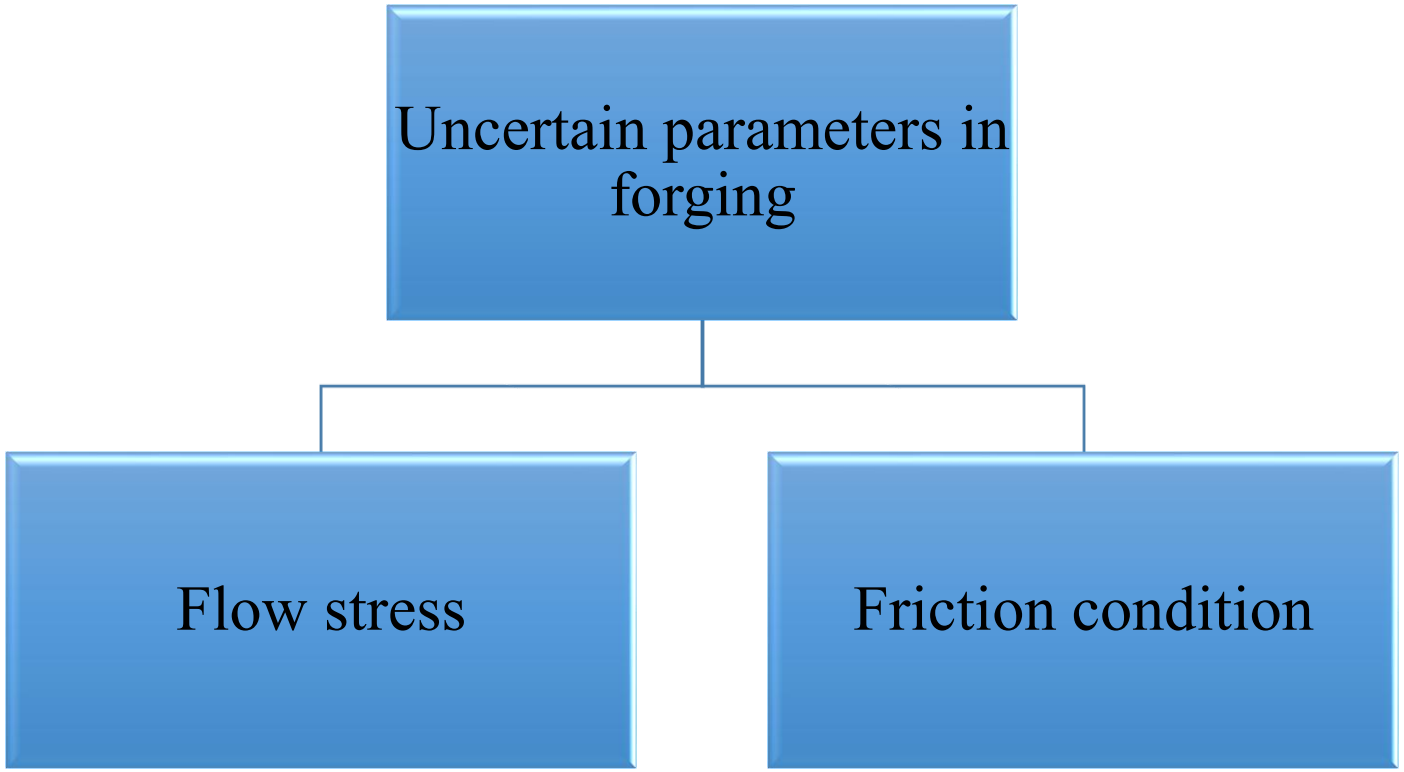
Need for modelling in metal forming

- Estimation of forging load and power requirement
- Pressure in the die and tools
- Stresses, strains and strain rate distributions in the material during processing
- Residual stress in the product
- Identifying defects
- Geometric accuracy
- Surface integrity
- Mechanical and metallurgical properties
- Microstructure

Challenges in modelling a metal forming process

Models	Challenges
Analytical	Assumptions, Exact knowledge of the material behaviour during deformation and Coulomb's coefficient of friction.
Numerical	Finite Element Method (FEM) simulations are computationally expensive for three-dimensional products.
Empirical	Requirement of large amount of data.

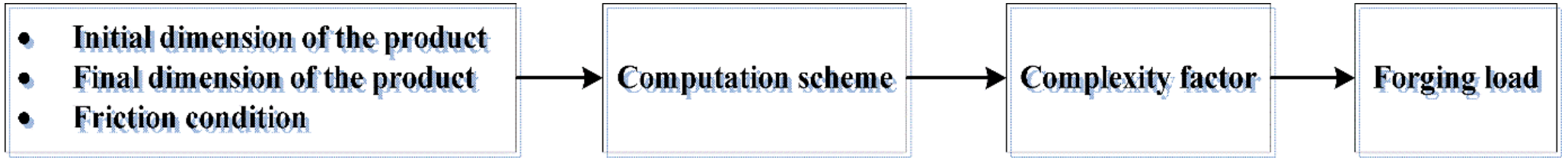
Uncertainty associated with metal forging



Data driven estimation of forging load

An empirical relation for the estimation of forging load was given long back (Altan and Fiorentino, 1971) as

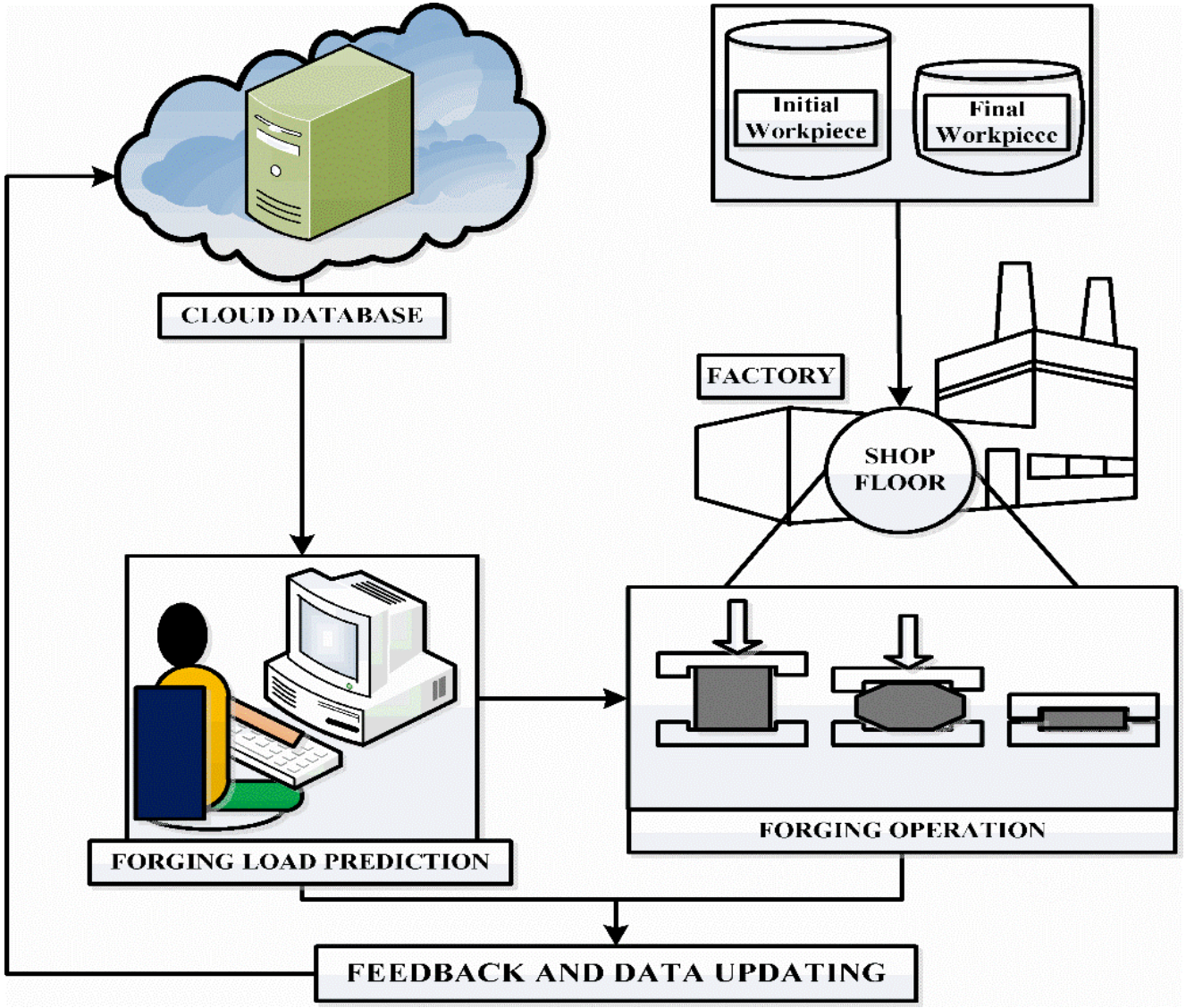
$$P = C_{\mathrm{p}} \sigma_o A_{\mathrm{p}}, \qquad (1)$$



Scheme for estimating forging load

The main challenge is to suggest a suitable value of (complexity factor) C_{p} such that it can be used for any forging case.

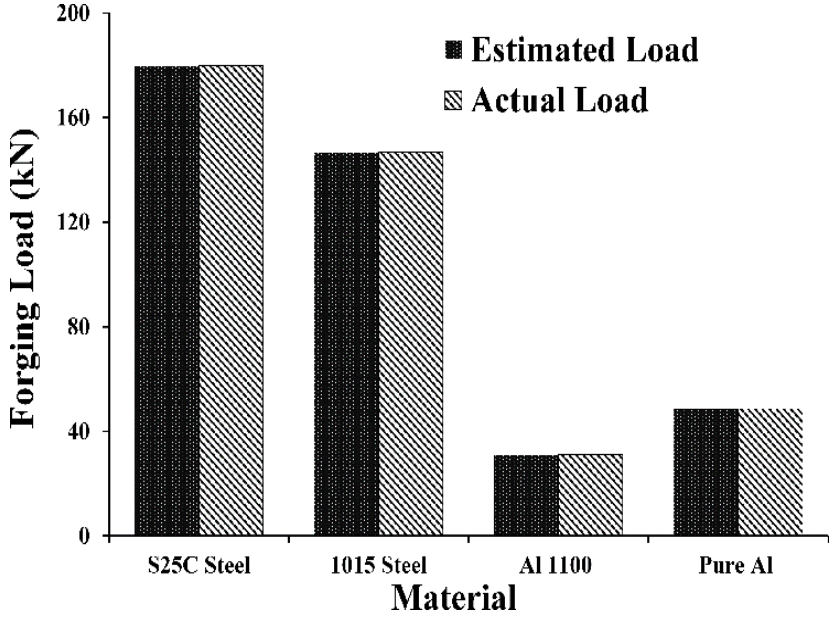
Estimating forging load using cloud database



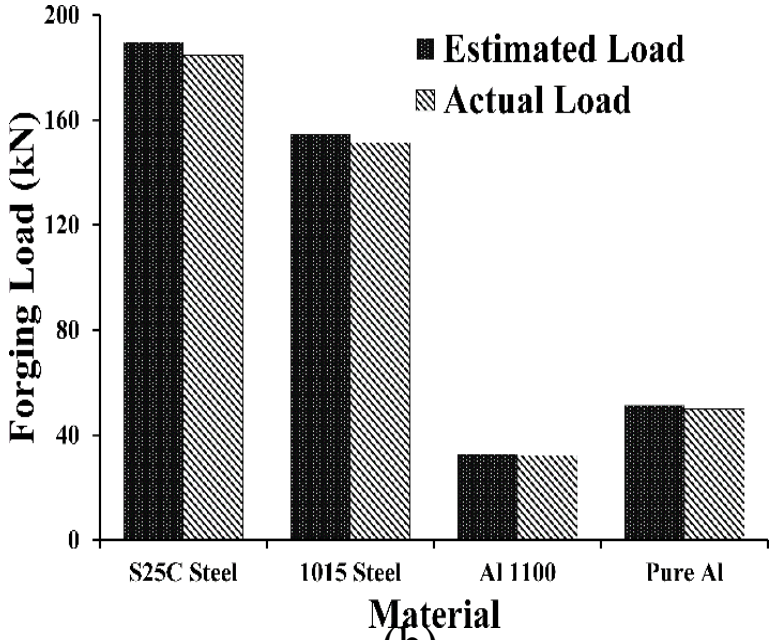
Estimating forging load in open die forging

From slab method the value of complexity factor is computed as

$$C_p = \frac{H}{\mu R^2}\left[\frac{H}{2\mu}\left\{\exp\left(\frac{2\mu R}{H}\right)-1\right\}-R\right]. \qquad (2)$$



(a)

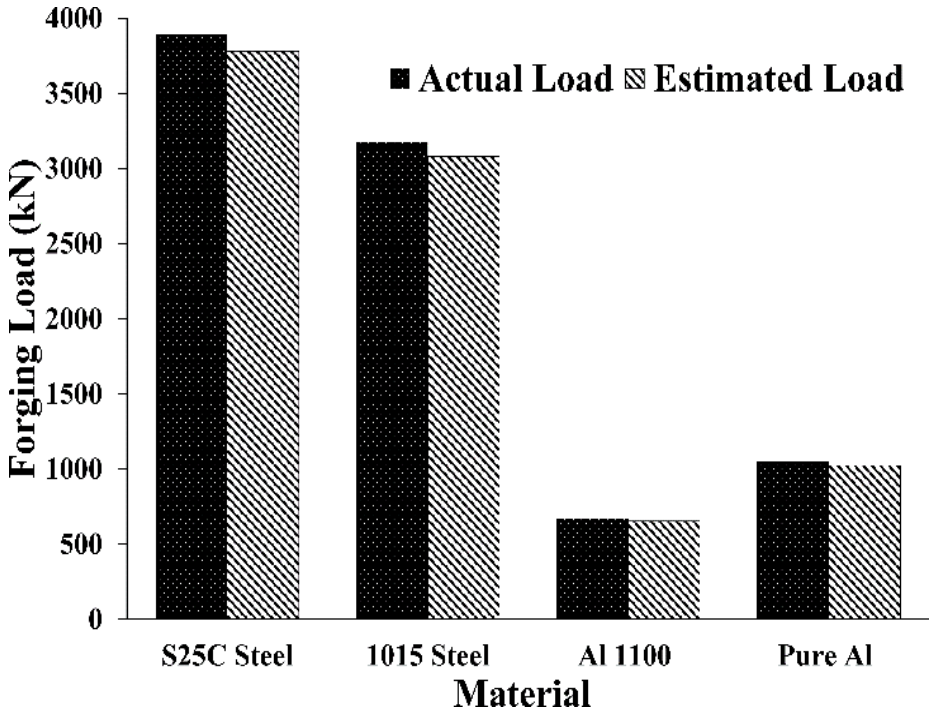


(b)

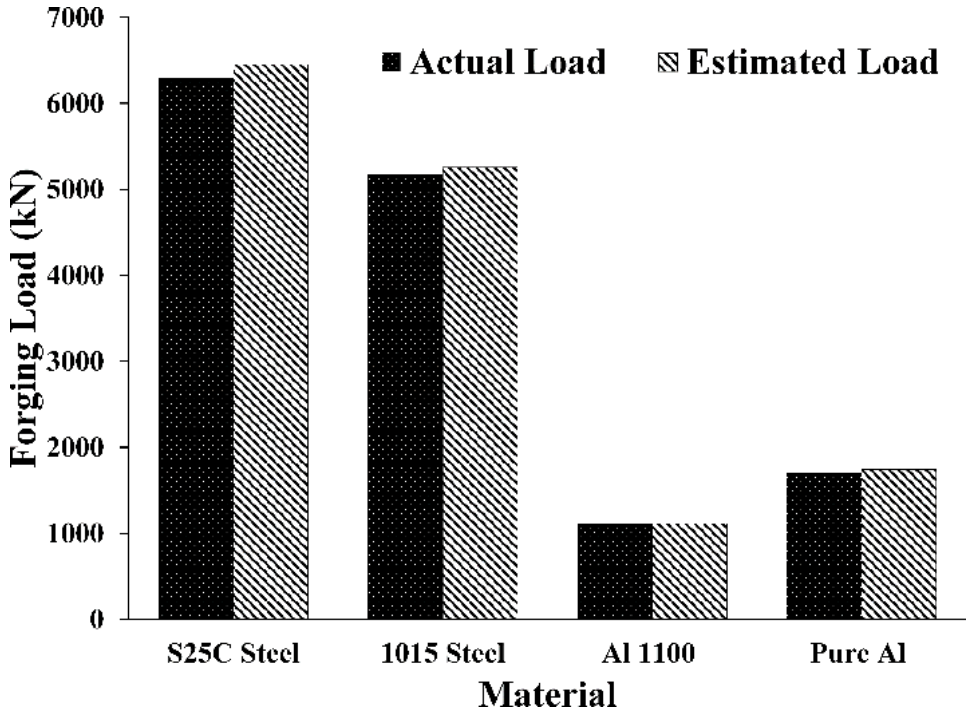
Estimation of forging load for 10% reduction (R = 10 mm and H = 20 mm) (a) lubricated condition, (b) non-lubricated condition

Chatterjee, K., Dixit, U.S., Zhang, J. and Petrov, P.A., 2022. A Methodology for Data-Driven Estimation of Forging Load. In *Recent Advances in Manufacturing Processes and Systems: Select Proceedings of RAM 2021* (pp. 11-22). Singapore: Springer Nature Singapore.

Estimating forging load in open die forging (cont.)



(a)

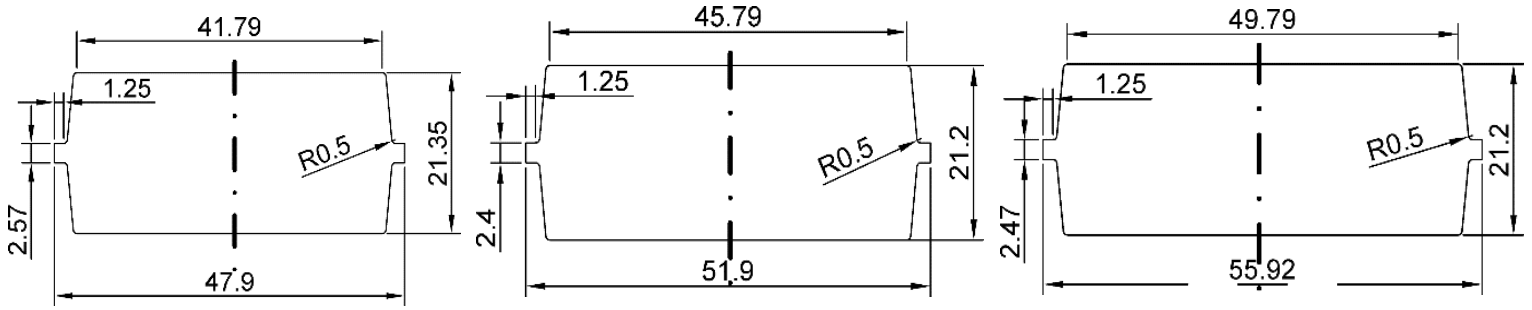


(b)

Estimation of forging load for 10% reduction (R = 40 mm and H = 10 mm) (a) lubricated condition, (b) non-lubricated condition

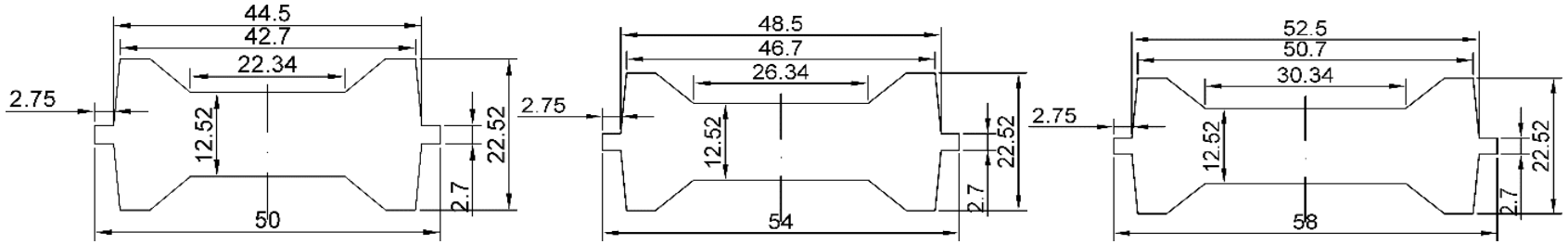
Estimating forging load in closed die forging

Forging load estimation for different sizes



(a) (b) (c)

Closed die forged Product 1: (a) Model 1, (b) Model 2 and (c) Model 3



(a) (b) (c)

Closed die forged Product 2: (a) Model 1, (b) Model 2 and (c) Model 3.

Estimating forging load in closed die forging (cont.)

The actual (virtually simulated) forging load for model 1 is 307.6 kN and the value of C_p is computed as 3.02.

Estimation of forging load for various sizes of Product 1

Model no.	Actual load (kN)	Estimated load (kN)	Error (%)	Base model
2	419.3	361.1	13.9	1
3	466.5	418.9	10.2	1
3	466.5	486.7	4.3	2

Estimation of forging load for various sizes of Product 2

Model no.	Actual load (kN)	Estimated load (kN)	Error (%)	Base model	Base product
1	347.5	363.7	4.6	1	1
2	363.6	405.3	11.5	1	2
3	422.1	467.6	10.7	1	2
3	422.1	419.5	0.6	2	2

Estimating forging load in closed die forging (cont.)

Estimation of forging load for different friction condition

Estimation of forging load for two friction conditions of product 1 and product 2

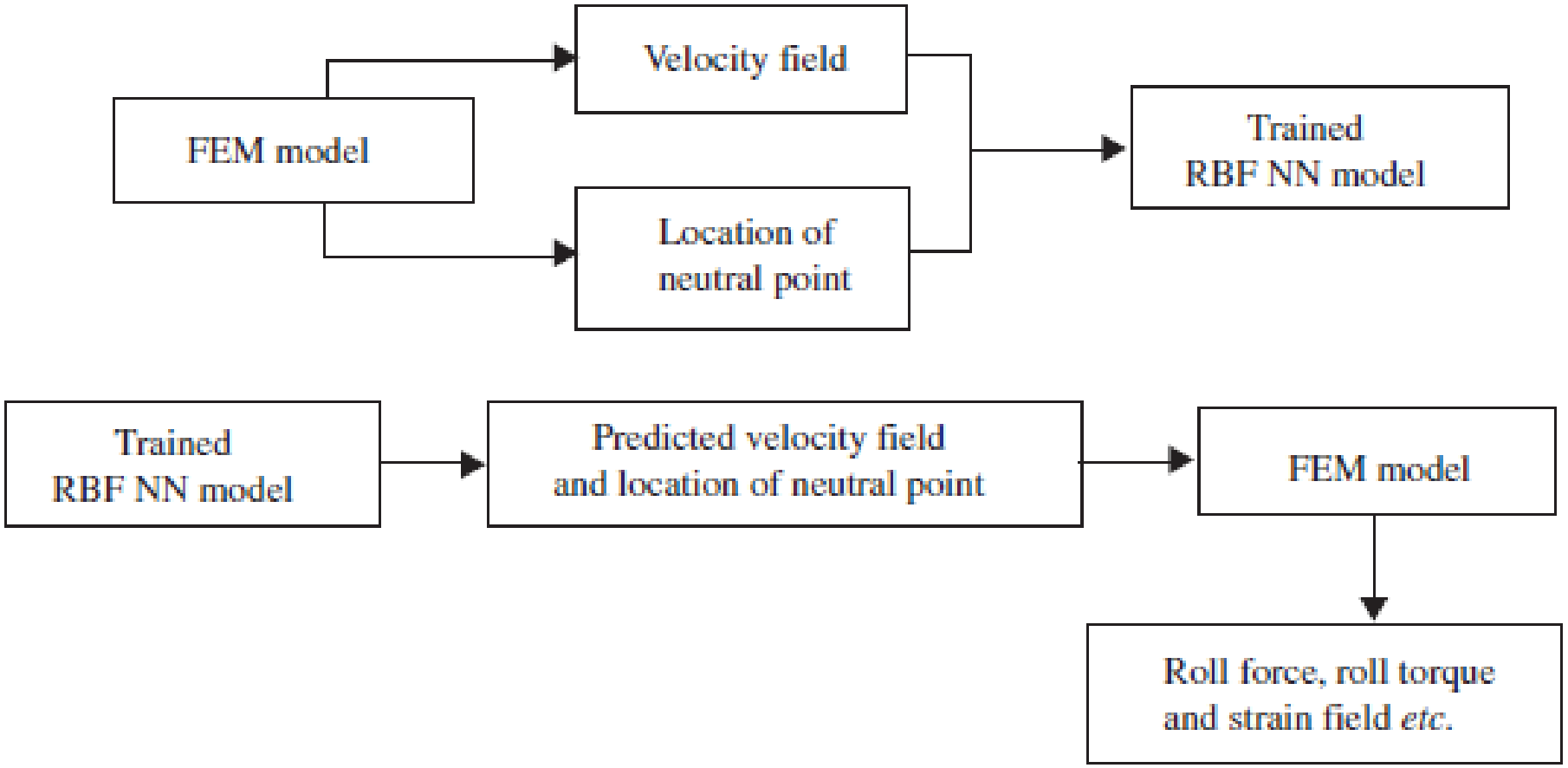
Product no.	Model no.	Friction condition	Actual load (kN)	Estimated load (kN)	Error (%)	Base model	Base product	Friction condition
1	1	non-lubricated	347.5	323.3	7	-	-	lubricated
2	3	non-lubricated	503.1	495.3	10.2	1	2	lubricated

Challenges associated with the complexity factor



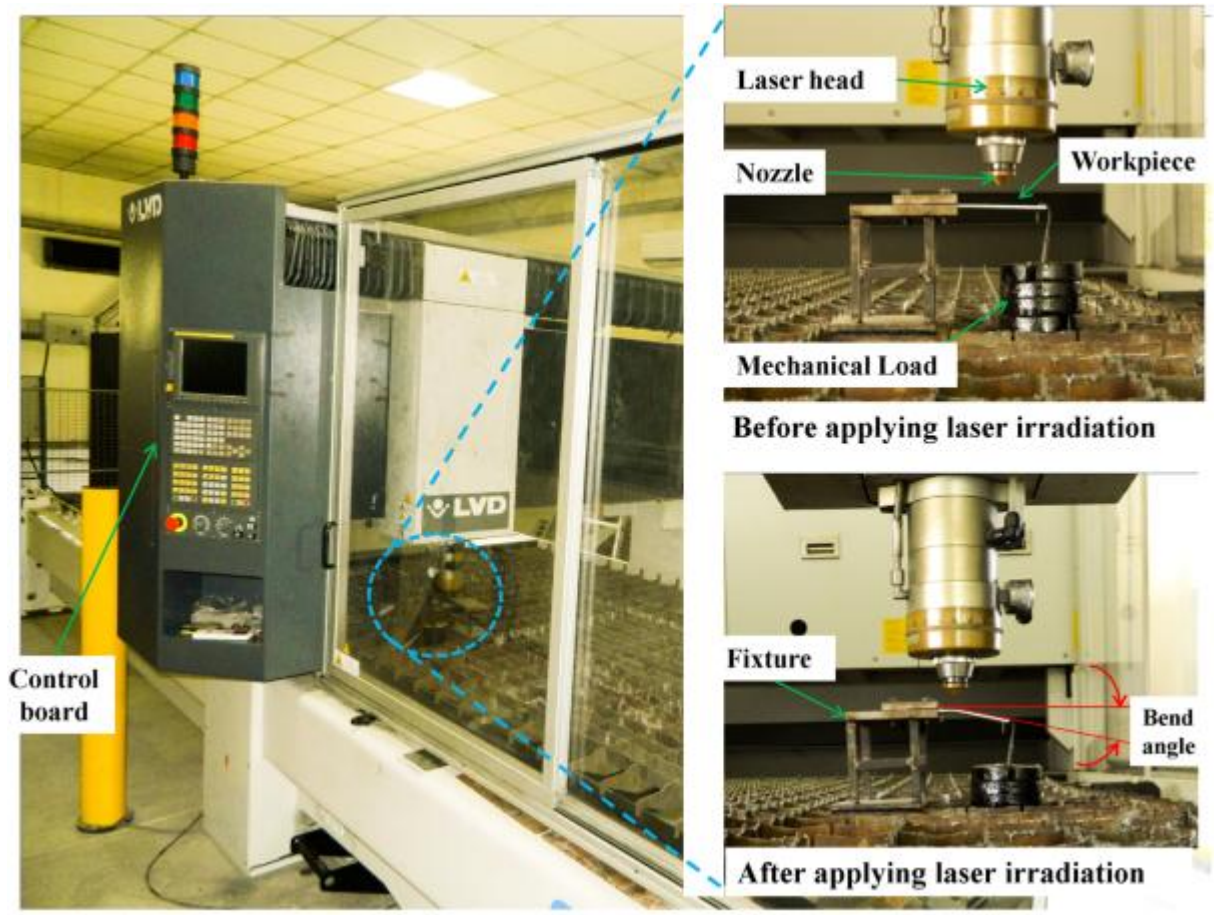
- Given any new shape, is it possible to suggest a suitable complexity factor in closed die forging for the accurate estimation of forging load?
- How to incorporate feature recognition for identifying similar products from the existing database?

Hybrid computational techniques in metal forming



FEM based neural network modelling for laser assisted bending

Fetene, B.N., Shufen, R. and Dixit, U.S., 2018. FEM-based neural network modelling of laser-assisted bending. Neural Computing and Applications, 29(6), pp.69-82.

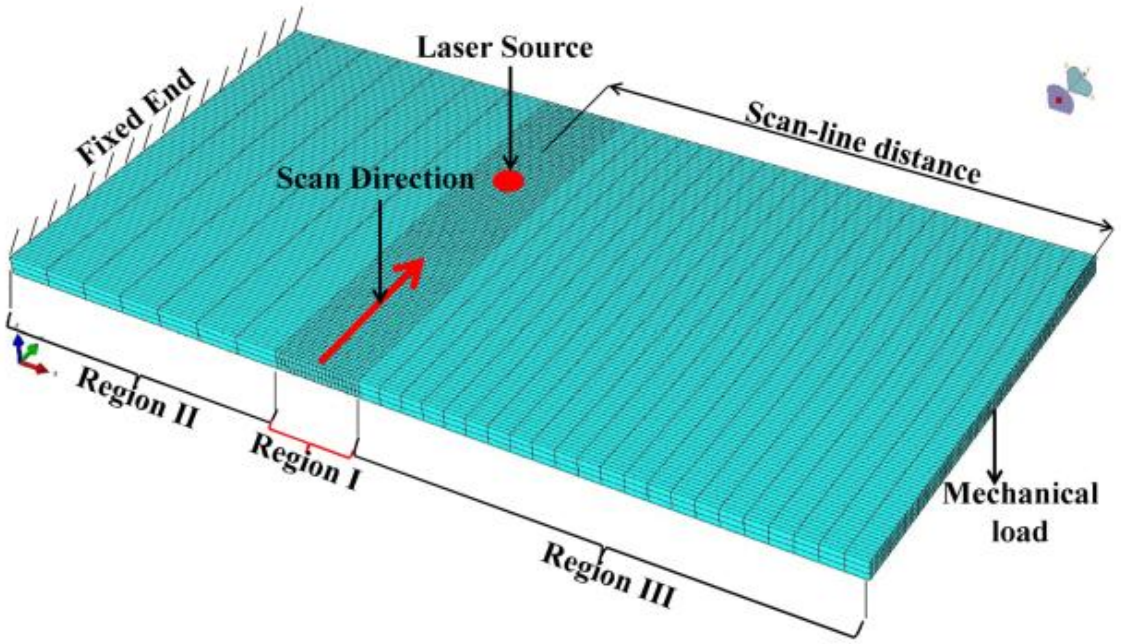


Experimental setup in the laser machine (Fetene et al., 2018)

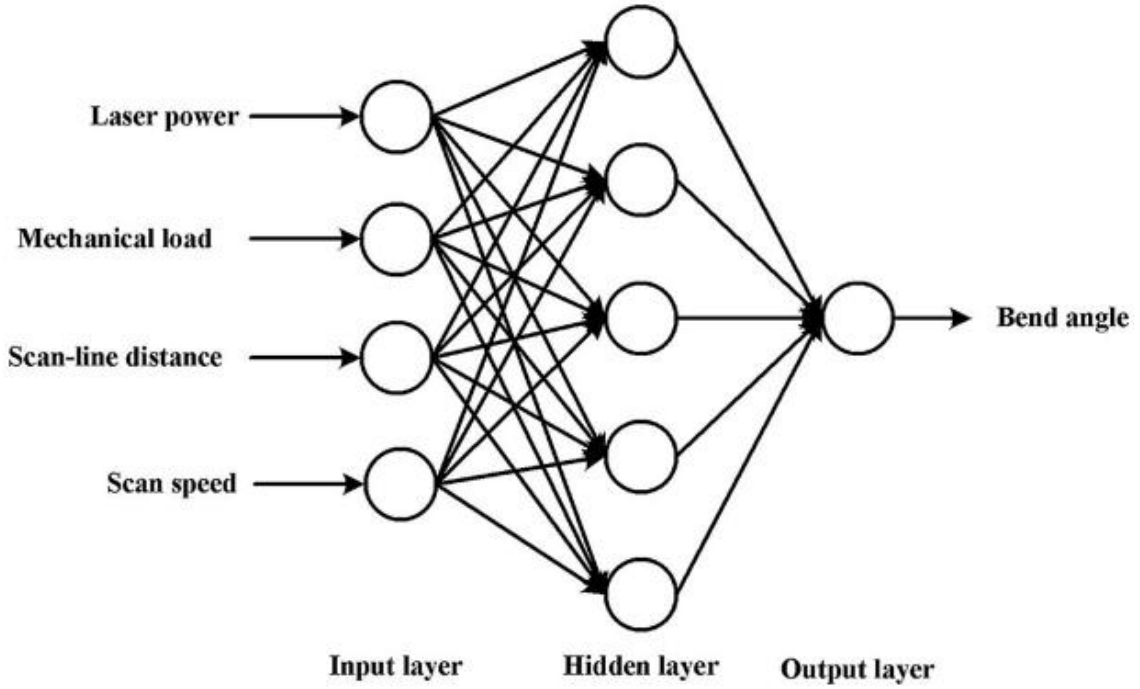
Physics-Informed Neural Networks

- Moseley, B., Markham, A., & Nissen-Meyer, T. (2023). Finite Basis Physics-Informed Neural Networks (FBPINNs): a scalable domain decomposition approach for solving differential equations. Advances in Computational Mathematics, 49(4), 62.

FEM based neural network modelling for laser assisted bending (cont.)

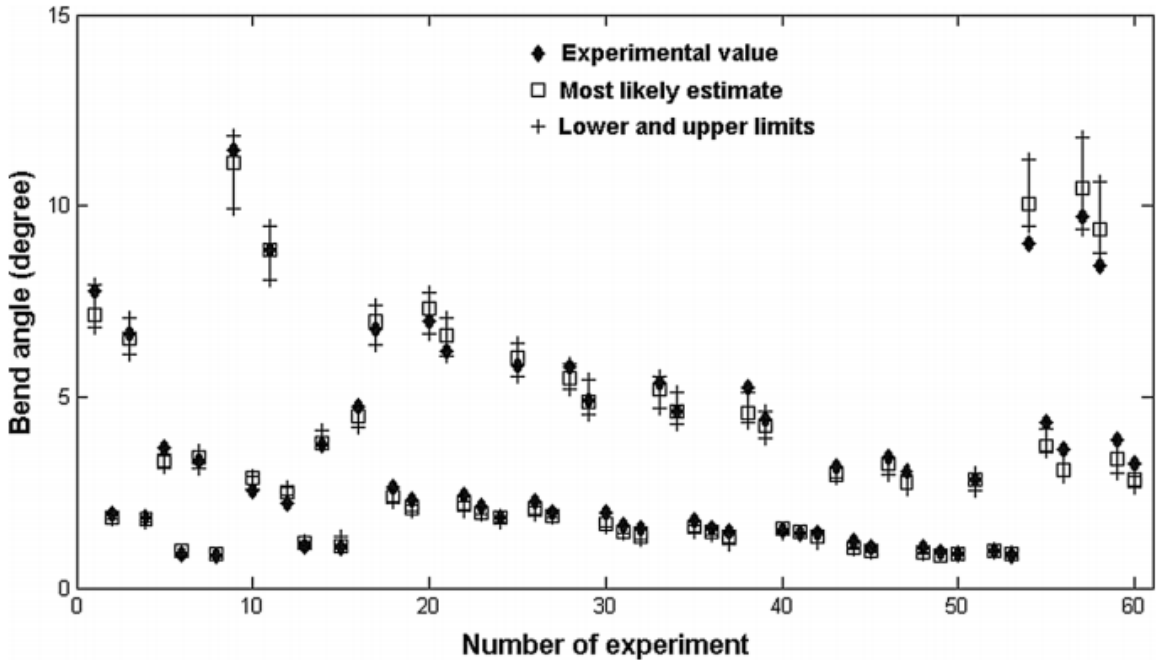


Experimental setup in the laser machine (Fetene et al., 2018)

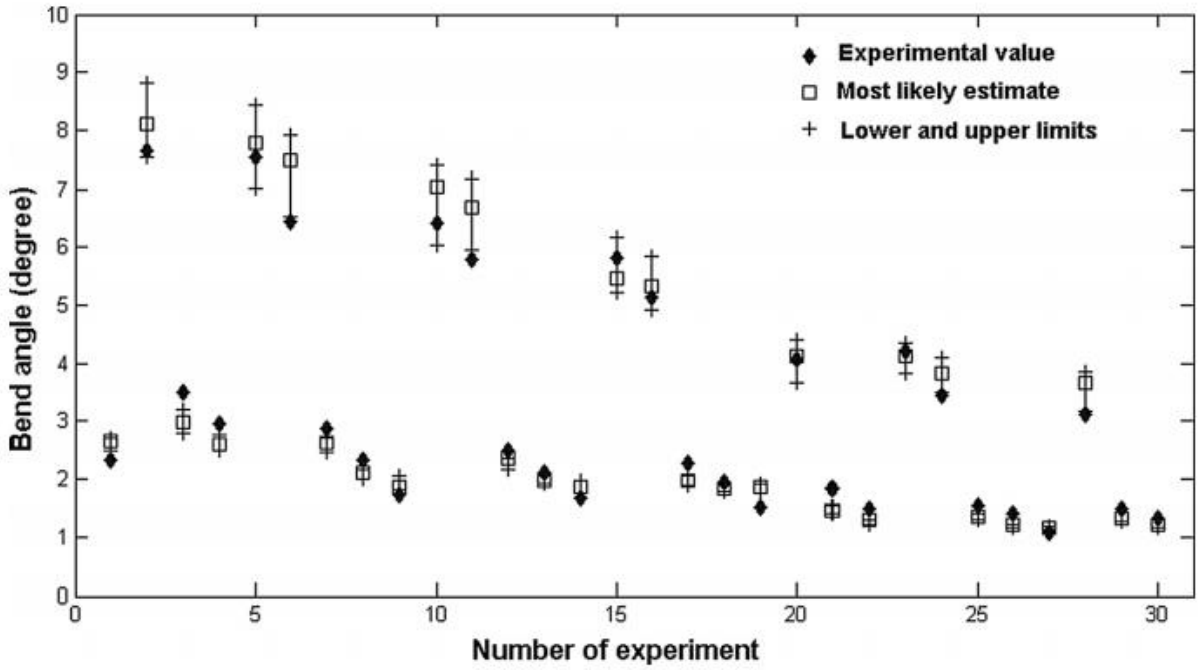


Multi-layer perceptron neural network architecture (Fetene et al., 2018)

FEM based neural network modelling for laser assisted bending (cont.)



Upper and lower bound estimation of the testing data (Fetene et al., 2018)

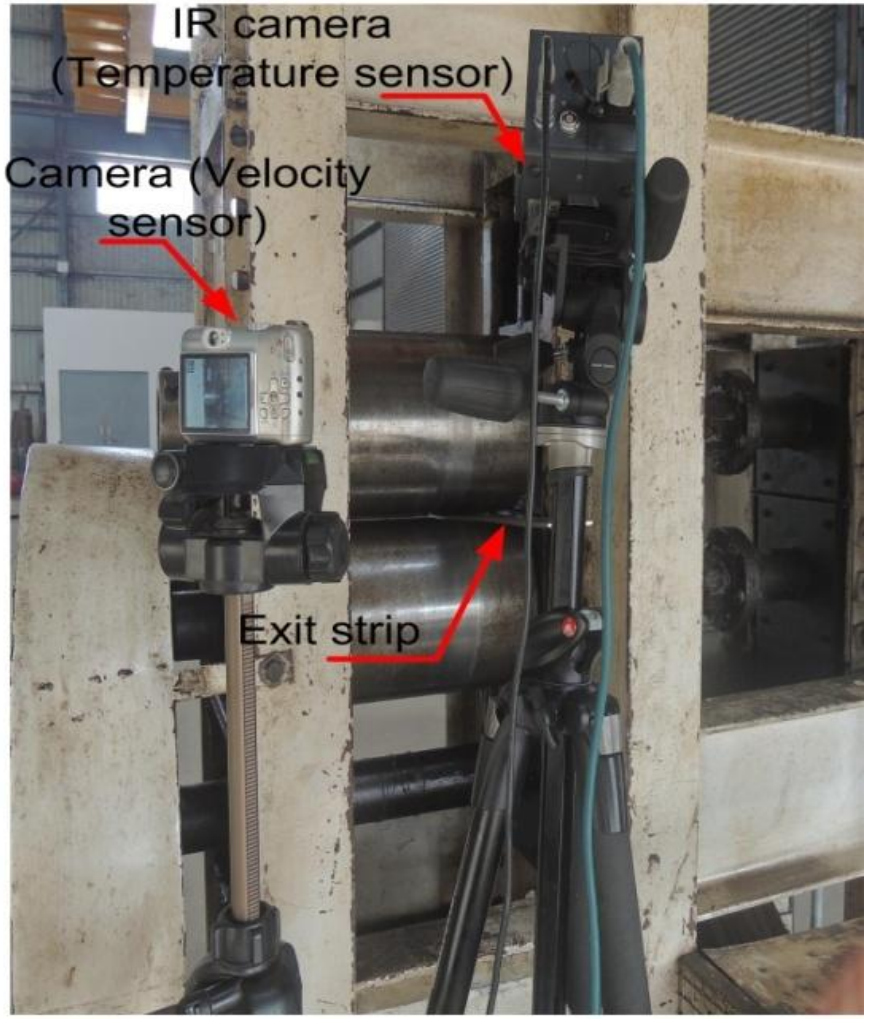


Upper and lower bound estimation of the validation data (Fetene et al., 2018)

Inverse Modelling for estimation of parameters



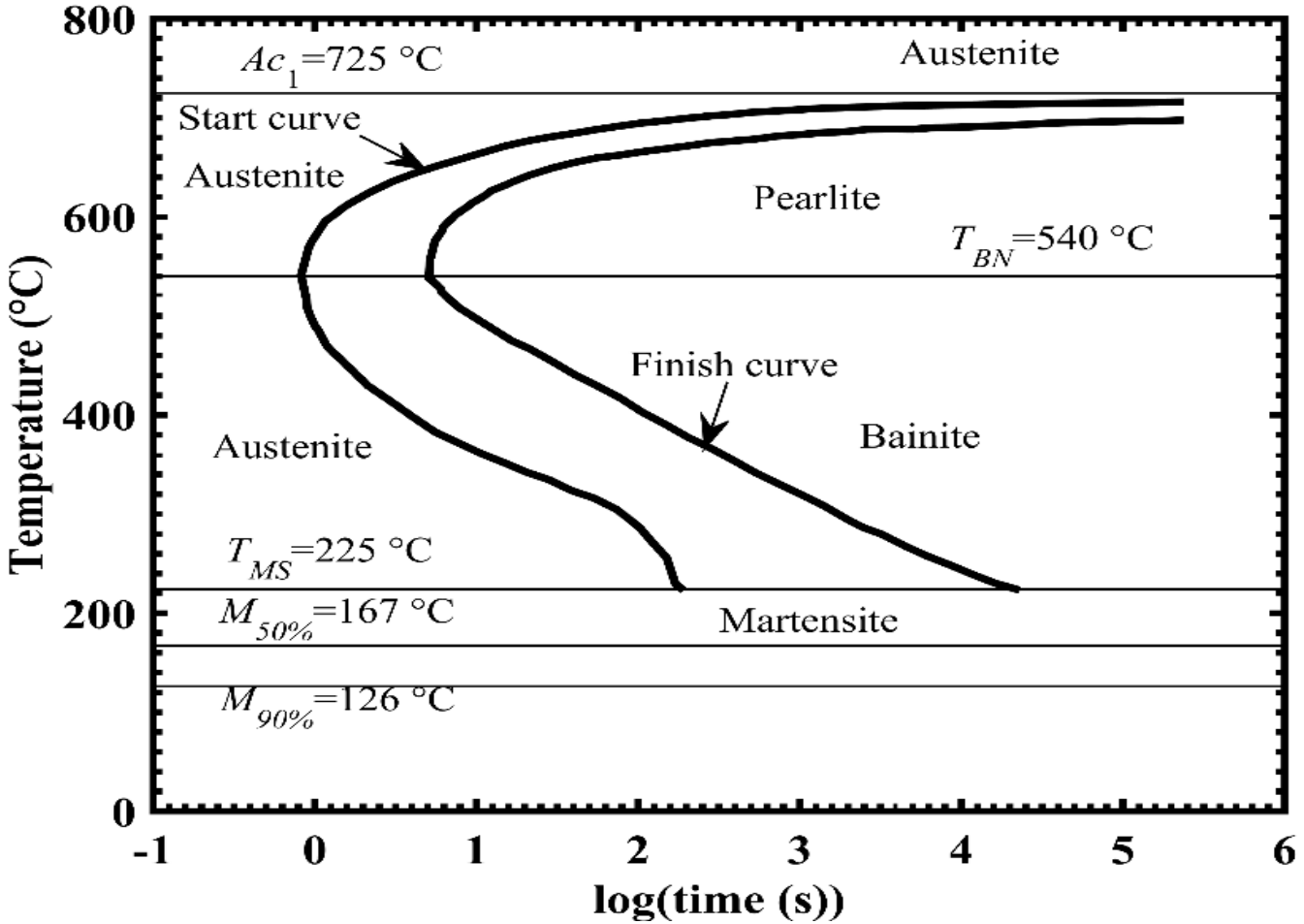
(a)



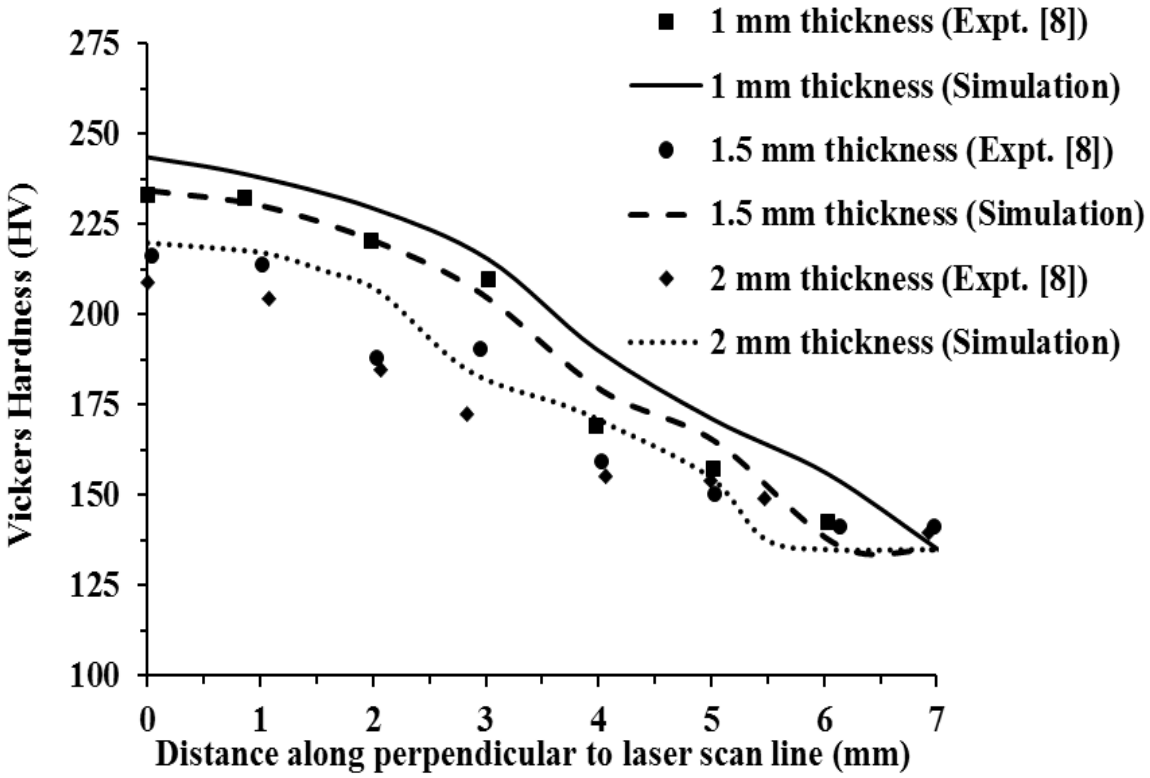
(b)

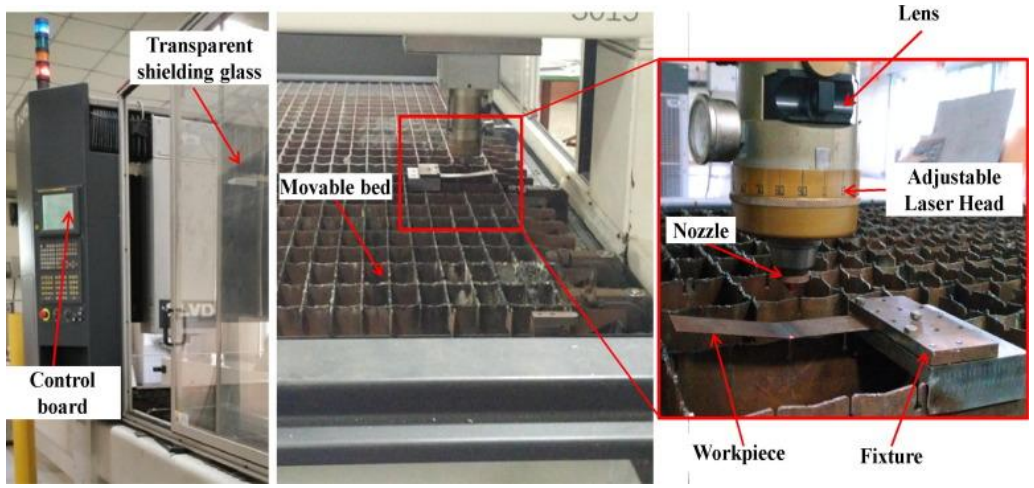
A laboratory rolling mill (a) front view and (b) arrangement for measuring the temperature and velocity of exit strip at the rear side

Phase transformation modelling



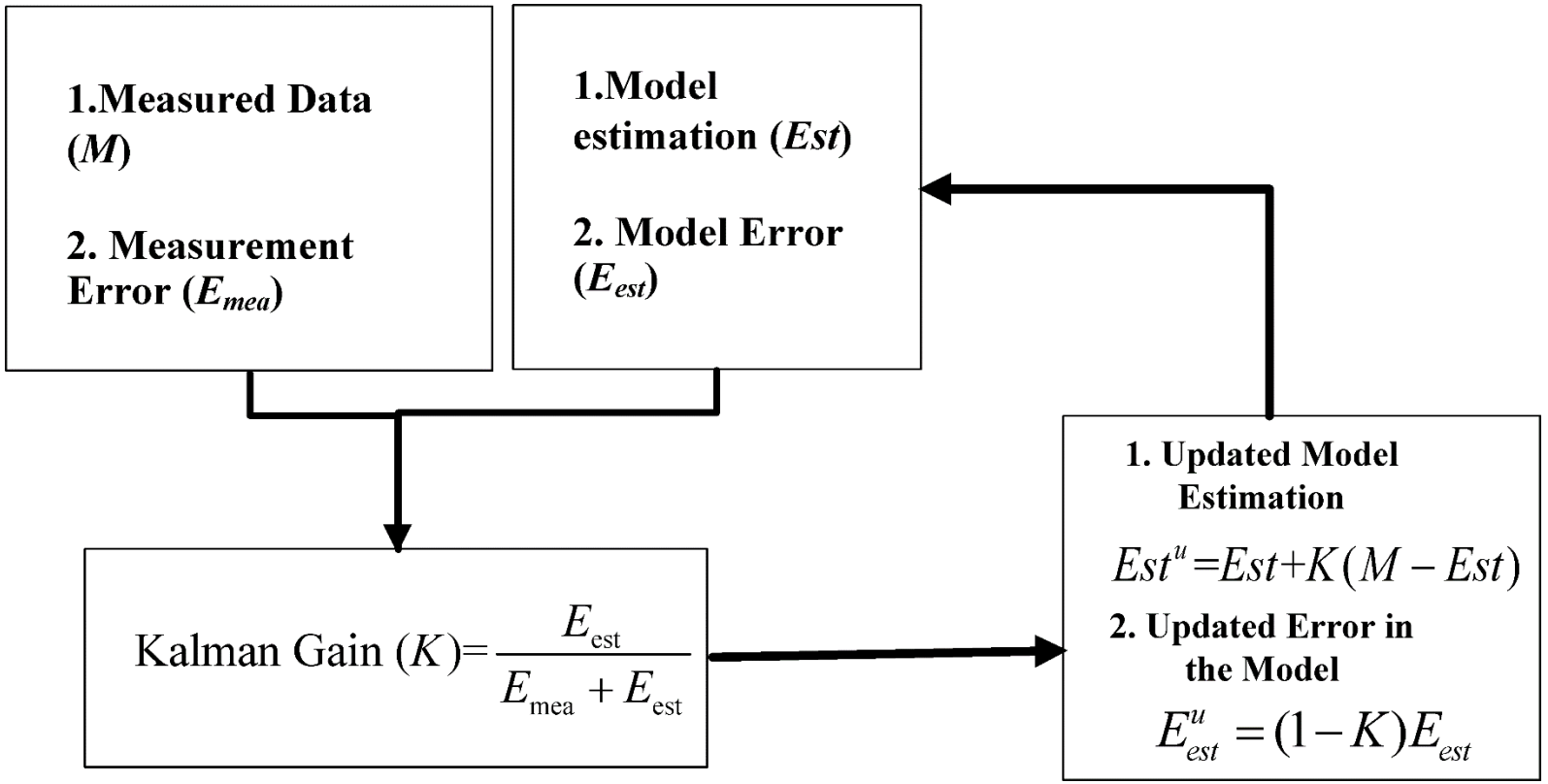
Hardness distribution in laser irradiated sheet





What is Kalman Filter?

It is a recursive estimator that uses the feedback from the previously estimated value to estimate the value of the current state from a set of noisy data (Barker et al. 1995)



Barker, A.L., Brown, D.E. and Martin, W.N., 1995. Bayesian estimation and the Kalman filter. Computers & Mathematics with Applications, 30(10), pp.55-77

Kalman Gain (K)

- Kalman Gain is a correction or weight factor for updating the current model estimation.
- The value of K can vary between 0 to 1.
- A value close to 1 indicates that the measurement data is more reliable than the value suggested by the mathematical model.
- A value close of 0 indicates an unreliable measurement data as compared to the value suggested by the mathematical model

Kalman Filter in Manufacturing

- Kalman filter was applied to measure the tool flank wear in milling operation (Niaki et al. 2015).
- Mechanistic modelling was used to suggest the current state estimation.
- The Kalman gain was evaluated based on the observed tool flank wear.
- The estimation suggested by the mechanistic model was then updated based on the calculated Kalman gain.
- The process continued till the desired level of accuracy was achieved.

- Kalman filter was then applied to measure the tool flank wear in a turning operation (Niaki et al. 2016).
- Mechanistic model was used to suggest the tool flank wear that was fine tuned by applying Kalman filter.

Niaki, F.A., Ulutan, D. and Mears, L., 2015. In-process tool flank wear estimation in machining gamma-prime strengthened alloys using kalman filter. *Procedia Manufacturing*, *1*, pp.696-707.

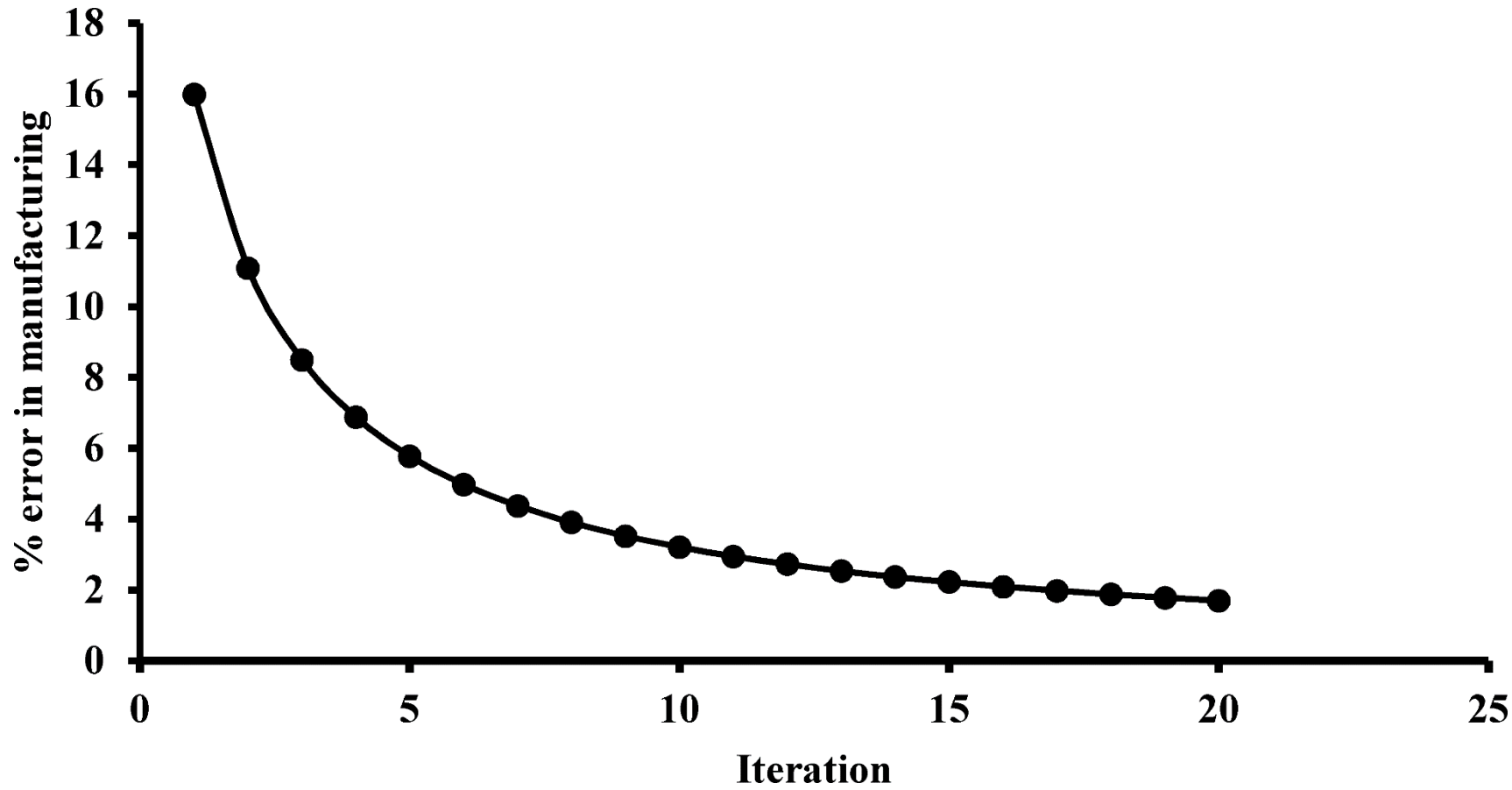
Niaki, F.A., Michel, M. and Mears, L., 2016. State of health monitoring in machining: Extended Kalman filter for tool wear assessment in turning of IN718 hard-to-machine alloy. *Journal of Manufacturing Processes*, *24*, pp.361-369.

Kalman Filter in Manufacturing

- Kalman filter has also been applied in the arena of metal forming.
- The concentricity measurement between large forgings and upsetting or piercing rod was carried out using the principle of laser ranging (Zhang et al. 2019).
- This type of measurement is crucial for maintaining the accuracy and precision of forgings.
- The distance captured by the laser range finders contained random noise further refined using the Kalman filter.

Zhang, Y., Wang, Y., Liu, Y., Lv, D., Fu, X., Zhang, Y. and Li, J., 2019. A concentricity measurement method for large forgings based on laser ranging principle. Measurement, 147, p.106838.

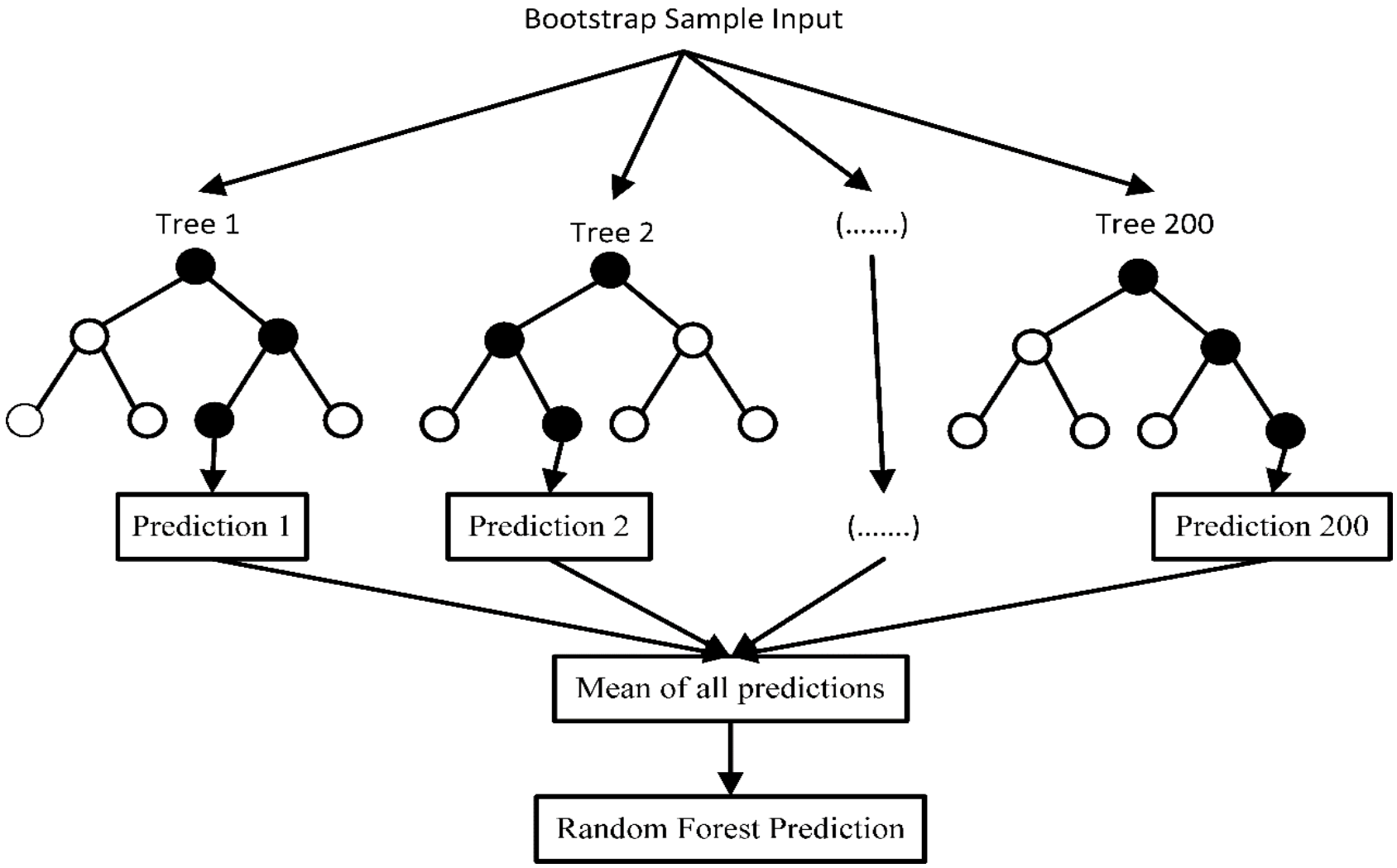
Kalman Filter in Manufacturing



Random Forest

- Random Forest falls into the category of ensemble learning.
- Ensemble learning is a technique that combines the predictions made by numerous machine learning algorithms to make it more accurate when compared to individual models.
- An Ensemble model therefore consists of numerous machine learning models.
- Training samples are created from the original dataset termed as Bootstrap sample.
- Bootstrap samples indicates sampling with replacement such that each data has an equal probability of it being selected, and each sample is the same size as the original training dataset.
- Ensemble prediction is defined as the one with the average of all the prediction being made.

Random Forest



The basic structure of Random Forest

Random Forest in Manufacturing

- Random forest was applied in a CNC milling operation to monitor the tool flank wear (Wu et al., 2017).
- Tool flank wear was monitored using the information collected from dynamometer, accelerometer and acoustic emission sensor.
- Random Forest was then applied to predict the health condition of a 3D printer.
- Wu et al. (2018) predicted the surface roughness in fused deposition modelling (FDM) products.
- Besides quality control and predictive maintenance, Random Forest can be applied to monitor the level of inventory in an industry and also towards supply chain management.

Wu, D., Jennings, C., Terpenny, J., Gao, R.X. and Kumara, S., 2017. A comparative study on machine learning algorithms for smart manufacturing: tool wear prediction using random forests. *Journal of Manufacturing Science and Engineering*, *139*(7).

Wu, D., Wei, Y. and Terpenny, J., 2018, June. Surface roughness prediction in additive manufacturing using machine learning. In *International Manufacturing Science and Engineering Conference* (Vol. 51371, p. V003T02A018). American Society of Mechanical Engineers.

Cyber Security in era of Industry 4.0

In the era of Industry 4.0, there is an increased usage of data among industries. Hence, there is an ever increasing risk of cyber attacks (Lezzi et al., 2018). The following are the reasons for developing a cyber security in the fourth industrial revolution:

- Industry 4.0 mainly incorporates data captured from various sensors that are used for data-driven modelling. Such data are vital for industries and need to be protected from cyber attack.

- Manufacturing processes often are centrally controlled. Any external access can cause disruption in the whole manufacturing process.

Encryption is a technique that help in preserving the data securely. Research is going towards enhanced protection of the data from cyber attacks.

Lezzi, M., Lazoi, M. and Corallo, A., 2018. Cybersecurity for Industry 4.0 in the current literature: A reference framework. *Computers in Industry*, *103*, pp.97-110.

Challenges of using ML for modern day industries

The following are few of the major challenges:

- Identifying outliers
- Applying appropriate ML algorithms for drawing inferences to any manufacturing data
- Cyber security associated while maintaining a data bank for applying ML algorithms
- Many ML algorithms (such as Gaussian process regression or support vector regression) are considered as a black box
- Hence, it is difficult to interpret the results obtained after applying such algorithms

Challenges for academic researchers

- The researchers have to put a lot of efforts in modelling and simulation, particularly in developing countries.
- As a number of companies already started selling digital twin applications, challenge is compounded.
- Should the researcher focus on developing their own applications or learn the appropriate application of commercial digital twin tools?
- A focus on in-house development of digital twin technology requires the strength to compete with bigger companies. On the other hand, too much reliance on commercial digital twin tools will pose threat to efficiency, effectiveness and security.
- It is best to bring this topic in the fold of manufacturing education and training


Thank you